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Volcanism and Short-Term Climatic Change in East Asian and World History, c. 1200–1699*

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There is . . . a tendency for the North Atlantic zone of main cyclonic activity to be shifted somewhat south in the summers after great [volcanic] eruptions, this accounting for many, perhaps most, of the coldest, wettest summers of the last three hundred years in western Europe and eastern North America. The same probably applies to the North Pacific (Lamb 1982: 59).

[A] cold summer with excessive prolonged rains is the commonest cause of a failure in the harvest [in the Tōhoku region of northern Japan]. . . . 1782–1787, 1833–1839, and 1866–69 are famous in Japanese historical writings as the worst famines in the last half of the Tokugawa period. . . . During [these] three great famines, . . . there were remarkably low temperatures in the summer months (Arakawa 1957b: 212).

Summer climates in Japan and China . . . are characterized by simultaneous variations of dry and wet areas. . . . In 1978, when it was extremely dry and hot in Japan and middle China, the axis of [the North Pacific High] was located northward, while in 1980, when it was

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exceedingly wet and cool in Japan and middle China, the axis was shifted far to the south (Mikami 1987: 69).

To those whose noon sunshine has been replaced by cimmerian darkness, the volcanic origin of their misery may be suffocatingly obvious if falling ash is sifting down around them. But it may be difficult for someone observing a small anomaly in his daily circumstances to relate that anomaly to events taking place on a volcano thousands of miles away, of whose very existence he may be unaware (Francis 1993: 368).

Part One: Some Recent Approaches to the Study of Volcanism and Short-Term Climatic Change

Introduction: The "Dust Veil" and "Volcanic Explosivity" Indexes

Although there has been serious discussion of the possible connection between volcanic activity and *short-term* climatic change for at least two centuries, research in this field has advanced greatly since the 1960s (see, for example, the summaries in Lamb 1988: 301–328; Rampino, Self, and Stothers 1988; American Geophysical Union 1992; Chester 1993: 158–185; Bradley and Jones 1995b; Robock 2000).[1] One reason for this advance is that after more than four decades of relative quiet following two massive eruptions in Alaska in 1912, there has been a substantial increase in both the frequency and magnitude of

[1] Perhaps the first person to suggest that connection was the American journalist and diplomat Benjamin Franklin, who did so in an extraordinary essay that was written in May 1784 and has been of great interest to modern researchers (Franklin 1882, VI: 455–457). Among the literally scores of volcanologists, atmospheric scientists, and other scholars who have quoted or otherwise cited Franklin's essay are: Humphreys 1929: 568–569; Lamb 1970: 433; Pollack et al. 1976: 1071; Bullard 1984: 511; Huggert 1991: 35; Rampino, Self, and Stothers 1988: 74; Schneider and Londer 1984: 277; Francis 1993: 369.

Here it should be noted that although some researchers have been interested in the possible connection between volcanism and "long-term" climatic changes (i.e., changes lasting for decades or even centuries), the focus here will be on much shorter periods. As H. Flohn has pointed out, "short-term, interannual fluctuations" of climate usually have a much greater impact on economic developments than "weak, long-term fluctuations" (Flohn 1981: 310–311). Unfortunately, as Flohn also notes elsewhere, "[t]extbooks on climatology contain hardly more than a few sentences on severe winters or dry summers, on droughts or floods. However . . . , the economic effects of such short-lived anomalies with a duration of a few months or years can be so negative or even destructive that meteorologists are not any longer allowed to neglect these events" (Flohn 1978: 311).

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global volcanism since the mid-1950s. Especially noteworthy in this increase were the July 1953 eruption of Alaska's Mt. Spurr, the March 1956 eruption of Mt. Bezymianny in Kamchatka, and the March 1963 eruption of Mt. Agung on the Indonesian island of Bali. This last eruption was particularly important because the large quantities of volcanic matter it injected high into the stratosphere helped to cause brilliant sunsets, hazy skies, and other optical effects that attracted the attention of observers from around the world for the next several years (Lamb 1970: 525; Newell 1981).

One of those observers was the pioneer meteorological historian H. H. Lamb, who was inspired to take up the complex question of volcanism and its possible impact on global climate. This resulted in a path-breaking article (Lamb 1970) that contained both a chronology of major eruptions over the previous 450 years and a preliminary assessment of their climatic impact. Particularly important for subsequent research was Lamb's compilation of a "Dust Veil Index" (DVI), an ingenious attempt to estimate both the quantities of fine ash and other matter injected into the atmosphere by major eruptions and how long they remained there. Using the relatively well-studied 1883 eruption of Indonesia's Mt. Krakatau (Krakatoa) as a benchmark, Lamb compiled this index by reviewing the records of more than 200 notable eruptions since 1500, taking into account both the qualitative comments of observers at the time and whatever quantitative information he could find on the volume of matter ejected, on post-eruption changes in solar radiation, and on subsequent fluctuations in global temperatures (Lamb 1970; Lamb 1972: 410–435; Grove 1988: 369). When his initial research was completed, Lamb estimated that between 1500 and 1699, volcanic dust veils dense enough to have had at least some short-term impact on climate (DVI $\geq$ 100) had been present over parts of the northern hemisphere in forty-two different years (Lamb 1970: 526–527; Appendix A).

Virtually as soon as it appeared, Lamb's DVI attracted the attention of scientists and historians alike. Prominent among the latter was John D. Post, who wrote the following in an important work published in 1977:

> Volcanic dust veils in the stratosphere will decrease the absorption of incoming radiation by reducing the transparency of the atmosphere, which in turn produces lower surface temperatures. The correlation of dense volcanic dust veils and intercontinental economic disturbances marked by high agricultural commodity prices can be documented . . .

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> for [a number of] historical periods. A partial list of the years when the two phenomena were present together . . . resembles a catalog of the pre-industrial crisis years, not only in Europe but often in North America and East Asia (Post 1977: xii).

To help illustrate Post's point about "pre-industrial" East Asia, it might be noted that the five worst famines to strike Japan during the relatively well-documented Tokugawa period (1603–1868)—the Kan'ei Famine (c. 1638–1643), the Kyōhō Famine (c. 1731–1733), the Hōreki Famine (c. 1755–1756), the Temmei Famine (c. 1783–1788), and the Tempō Famine (c. 1832–1838)—*all* coincided with at least one, and usually several, years to which Lamb assigned northern hemisphere DVI ratings ≥ 100 (Nakajima 1976: 8–10; Arakawa and Usami 1985: 324–325; Lamb 1970: 526–527). Although China's much greater size and agricultural diversity make such "national" generalizations more difficult, recent research indicates that over the five centuries between 1470 and 1969, the five worst years of consecutive drought in China *as a whole* occurred between 1637 and 1641. Lamb lists all five as dust veil years in the northern hemisphere, three of them (1638, 1640, 1641) with DVI ratings ≥ 100 (Zhang and Lin 1992: 291–292; Lamb 1970: 526–527; Appendix A). As will be discussed in Part Two below, the food shortages and economic dislocations that occurred in China during the late 1630s and early 1640s not only contributed directly to the fall of the Ming dynasty in 1644 but also coincided with sharp climatic fluctuations, severe agricultural problems, and significant political unrest in Japan, Korea, Manchuria, and many other parts of the world (Parker and Smith 1997; Marks 1998: 134–162, 195–210).

Although Lamb's early publications on volcanism and climate are widely and properly credited with "laying the foundations" for serious modern scholarship on the subject, other researchers soon began to express reservations about some of the methods he had used to compile the DVI. For example, uncomfortable with Lamb's use of post-eruption temperature changes to estimate the magnitude of volcanic activity, Newhall and Self developed an alternative method for classifying the size and power of past eruptions, one that was based primarily on volcanological criteria (Newhall and Self 1982; see also Chester 1993: 175–178; Simkin and Siebert 1994: 23–25; Bradley and Jones 1995b: 610–613). The result was what they called the "Volcanic Explosivity Index" (VEI), an estimate of an eruption's "explosivity" based on various combinations of its magnitude, intensity, destructiveness, dispersive power, violence, and energy release (Newhall and Self 1982: 1234; Simkin and Siebert 1994: 23). Constructing a 9-point (0–8) scale

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somewhat similar to the Richter scale used for measuring earthquakes, Newhall and Self investigated more than 8000 historic and prehistoric eruptions, providing "explosivity" (VEI) ratings for 6295.[2] Of these, 699 were rated at VEI 0 ("non-explosive"), 845 at VEI 1 ("small" or "gentle"), 3477 at VEI 2 ("moderate" or "explosive"), 869 at VEI 3 ("moderate-large" or "severe"), 278 at VEI 4 ("large" or "cataclysmic"), 84 at VEI 5 ("very large" or "paroxysmal"), 39 at VEI 6 ("colossal"), 4 at VEI 7 ("super-colossal"), and 0 at VEI 8 ("mega-colossal"). No fewer than forty-seven firmly dated eruptions rated at VEI 4–7 are thought to have occurred in various parts of the world between A.D. 1200 and 1699 (Simkin and Siebert 1994: 23, 179–251; Appendix B).

Ice-Core Studies

Although the VEI has proven to be an extremely useful tool in establishing the magnitude of many past eruptions, it, like the DVI before it, has some important limitations. First, the documentary record of volcanic activity on which these and related chronologies partially are based is far from complete. Before the late fifteenth century when the European "voyages of discovery" began to open up many parts of the world to systematic record keeping for the first time, written accounts of individual eruptions apparently existed for only about seventy volcanoes around the world, most of which were in the Mediterranean region, Iceland, or Japan (Simkin and Siebert 1994: 26). Gaps such as this in the historical record naturally make it difficult to generalize with confidence about the impact of volcanism on global climate even for relatively recent times (Simkin 1994: 913; Bradley and Jones 1995b: 618).[3] However, since the 1970s, and particularly since the early 1980s, scientists have had at their disposal a vast array of new and improved instruments, some of which are housed in orbiting satellites and specially equipped aircraft, which, for the first time, can monitor many aspects of volcanic activity as it is occurring (Hofmann and Rosen 1977, 1982; McPeters, Heath, and Schlesinger 1984). Recent research done with these instruments demonstrates that, as Lamb and others had acknowledged in their earlier research but did not then have

[2] These eruptions, some with revised VEI ratings, have been included in Simkin and Siebert 1994: 179–251.

[3] As Jean M. Grove has noted, "Astronauts now report eruptions seen from spacecraft and from their counts it has been estimated that one-quarter to one-third of the eruptions that occur have not been brought to the attention of the Smithsonian [Institution's Global Volcanism Program]" (Grove 1988: 374).

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the equipment to measure accurately, the chemical composition of volcanic emissions is of crucial importance in understanding the relationship between volcanism and climatic change:

> Large eruptions that are able to inject sulphur into the stratosphere can lead to a change in the radiation balance and temperature of the earth. Such a climate "forcing" by volcanic eruptions appears to be one of the most significant short-term changes imposed by nature. Recent major eruptions have caused the surface temperature to fall globally by 0.2–0.3° C for one-to-three years. *Even such a relatively minor drop could, however, translate into more severe local or regional temperature changes. Historically, such downturns in temperature have had catastrophic consequences for the world's food supply* (American Geophysical Union 1992: 3–4 [my emphasis]; see also Rampino and Self 1984; Simkin and Siebert 1994: 24).

Although not all of the physical or chemical processes involved in this kind of climate "forcing" are completely understood, certain types of eruptions inject significant quantities of water vapor, sulfur dioxide, hydrochloric acid, and mineral ash high into the atmosphere. Because hydrochloric acid dissolves in water, it usually is rained out of the atmosphere fairly quickly. The sulfur dioxide, on the other hand, remains in a gaseous state from which it eventually is converted into sulfuric acid. This acid, in turn, condenses into a "fine mist" of sulfate aerosols which helps to deflect the sun's radiation back into space and cool the earth's surface below (American Geophysical Union 1992: 5).[4] However, as the earth cools, it does so unevenly, a fact that can have a significant impact on atmospheric circulation and thus on global wind and precipitation patterns (Wexler 1952, 1956; Handler 1989). In part because the volcanic aerosols that reach the stratosphere initially are concentrated in a relatively narrow band of latitudes, those areas of the earth that are directly beneath the "aerosol cloud" usually cool more quickly than areas that are more exposed to solar radiation. Because of this, and because land areas and oceans cool at different rates, volcanic aerosols can help to produce "climatic anomalies" on a global scale (Handler 1989: 233). For example, during the summer of 1954 when there was considerable "turbidity" in the atmosphere over the northern hemisphere, at least some of which was caused by "sulfur-rich" volcanic activity,

[4] These sulfate aerosols also sometimes serve as nuclei for cirrus clouds, thus helping to increase overall cloudiness and further reduce the amount of solar radiation reaching the surface of the earth (American Geophysical Union 1992: 5; Akin 1991: 55–56).

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> Great Britain and central Europe experienced cold and stormy weather, and many floods were reported from Iran, East Pakistan, and eastern India. North Japan also was dominated by abnormal coolness and persistent rain for nearly two months, and in China the Yangtze Valley was flooded for hundreds of kilometers, while the central part of the United States was plagued by hot weather and drought (Wada 1978: 398; see also Yamamoto, Iwashima, and Hoshiai 1978; Newhall and Self 1982: 1236; Suda 1955; Lamb 1970: 466; Flohn 1981: 315).[5]

Given these facts, it seems clear that any serious analysis of the connections between volcanism and climatic change must take into consideration the amount of sulfur dioxide that is introduced into the atmosphere by a given eruption or series of eruptions. For example, modern technology has shown that some quite powerful eruptions, such as that at Mt. St. Helens in the western United States in 1980 (VEI 5), are "sulfur-poor" and thus have a less significant impact on *global* climate (Robock 1981b).[6] However, other eruptions, such as those at El Chichón in southern Mexico in 1982 (VEI 5) and at Mt. Pinatubo in the Philippines in 1991 (VEI 6), are rich in sulfur and thus are much more likely to alter global climatic conditions in a major way (Robock 1983; Rampino and Self 1984; Francis 1993: 382–385; see also Matson and Robock 1984; McCormick 1992; Appendix C).[7] Therefore, it clearly is important to try to distinguish between sulfur-rich and sulfur-poor eruptions and to establish as precisely as possible when there was a significant "loading" of sulfate aerosols in the atmosphere. It is in this important area of research that work done by C. U. Hammer and other scholars at the Geophysical Isotope Laboratory of the University of Copenhagen has been particularly useful.

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[5] Drawing on the research of H. Arakawa, K. Tsutsumi, and others, Lamb notes that in addition to a significant eruption at Alaska's Mt. Spurr in July 1953, global climate in 1954 also may have been affected by a series of atomic bomb tests in the atmosphere carried out by the United States and the Soviet Union from late 1952 to early 1954 (Lamb 1970: 466).

[6] As has recently been pointed out, however, "*regional variations* in the atmospheric response to volcanic eruptions" also must be taken into consideration (American Geophysical Union 1992: 3–4 [my emphasis]; see also Crowley et al. 1993: 211). Although the *global* impact of the May 1980 eruption at Mt. St. Helens may have been "negligible" (Francis 1993: 382), it nevertheless is important to note that the summer of 1980 saw unusually cool temperatures, persistent rainfall, and poor growing seasons in much of central China, Korea, Japan, the Ukraine, and western Russia. Europe also experienced cool and rainy weather during the summer of 1980, although grain production there did not suffer as badly as in some other parts of the world (Mikami 1987: 69; Kondo 1988: 784–785; Le Comte 1981).

[7] To further complicate matters, Handler has argued that some eruptions with VEI ratings of 2 or 3 may have released enough sulfur dioxide into the atmosphere to have had "a significant impact on climate . . . " (Handler 1989: 234).

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Beginning in the late 1970s, Hammer and his colleagues published a series of articles (Hammer 1977; Hammer et al. 1978; Hammer, Clausen, and Dansgaard 1980, 1981) that explained the ways in which ice cores could be used to date and analyze certain kinds of volcanic activity. The importance of this research, which has been continued and expanded by Hammer and others (see, for example, Thompson et al. 1992; Zielinski et al. 1994; Thompson 1995; Robock and Free 1995, 1996a, 1996b; Clausen et al. 1997; de Silva and Zielinski 1998), lies in the fact that the "fallout" from sulfur-rich eruptions sometimes leaves a sulfate-rich ice layer, which is detected as an acid or sulfate "spike" in ice cores around the world. Moreover, because those spikes often can be dated with some precision, it sometimes is possible to determine approximately when the sulfate loading of the atmosphere is likely to have had a significant impact on global climate and when there may have been climatically significant eruptions that either have been poorly dated or do not appear elsewhere in the historical or scientific record (Palais, Germani, and Zielinski 1992; Zielinski et al. 1994).

Recent analysis of eight northern hemisphere ice cores by Robock and Free has revealed that in the 500 years between 1200 and 1699, there appear to have been at least 110 years in which significant sulfate acid spikes were deposited in two or more locations (Robock and Free 1996a; Appendix C). However, it should be emphasized that because it sometimes takes several years for volcanic aerosols produced in the lower latitudes to reach the higher latitudes from which most ice cores are taken, some sulfate acid spikes may be a "lagging indicator" of when a given eruption, or a closely spaced series of eruptions, is likely to have had a climatic impact on key agricultural regions around the globe. To cite one example that will be discussed in Part Two below, the powerful and "sulfur-rich" eruptions at Japan's Mt. Komagatake (VEI 5) in 1640, at Indonesia's Mt. Kelut (VEI 4?) in 1640, and at either Indonesia's Mt. Awu (VEI 5?) or the Philippine's Mt. Parker (VEI 6) in 1641, apparently deposited their most significant sulfate acid spikes in the winter of 1641–42, which was six to twenty-four months after those eruptions already had begun to have a noticeable impact on global climate.

Tree-Ring Studies

As useful as this ice-core research has proven to be, it unfortunately cannot provide a complete record of sulfur-rich eruptions. For one thing, most of the ice cores that have been studied to date are taken from "high latitude" (60°–90° N or S) sites in Greenland and Antarc-

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tica and thus are "strongly biased" toward high latitude eruptions. Moreover, the acid spikes themselves

> may differ significantly in magnitude from one ice core to another and *important [volcanic] events may not appear at all in some ice core records*. This reflects the fact that the spatial distribution of acidic snowfall varies after volcanic eruptions and hence the record or deposition will vary from one ice coring site to another. [Finally], in some low latitude ice cores, deposition of alkaline aerosols may neutralize the volcanic acids and hence eliminate the eruption signal (Bradley and Jones 1995b: 615 [my emphasis]).

For these and other reasons, scientists recently have looked to other techniques to help fill in the gaps in the ice-core record of volcanic activity. One such technique involves dendrochronology, the study of annual tree-ring growth to extract information on climate and climatic change. Particularly important for our purposes here is the fact that a number of "frost-damage zones" or "frost rings" have been found in subalpine bristlecone pines in the western United States. Such frost damage is a rare phenomenon in mature trees, one caused by temperatures that drop to well below freezing at some point during the summer growing season. In the western United States, where the largest number of accurately dated bristlecone pines are located, such "anomalous" drops in summer temperatures often appear to be linked to volcanically induced changes in atmospheric circulation, changes that include a

> southerly displacement of the general westerly [wind] flow and/or more frequent development of an upper level trough, accompanied by occasional outbreaks of unseasonably cold air from higher latitudes. . . . Such a scenario seems to have been followed in the frost-ring year of 1884, where an examination of daily surface pressure maps for 9 and 10 September [1884] shows a very large high-pressure area extending from northern Saskatchewan and Manitoba down through California, Nevada, and Utah, probably representing an outbreak of cold, Arctic air that took place unusually early, near the end of an already cool and delayed growing season (LaMarche and Hirschboeck 1984: 124; see also Lamb 1982: 34–35).

Referring to the 1884 frost damage as the "Krakatoa effect" after the "colossal" (VEI 6) and sulfur-rich eruption of Mt. Krakatau in August 1883, LaMarche and Hirschboeck go on to point out that most incidents of frost damage to bristlecone pines in the western United States can be dated confidently to years when there is known to have been a significant loading of volcanic aerosols in the atmosphere, thus

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helping to confirm the information found in some ice-core records. Equally important, however, is the fact that frost damage in bristlecone pines also has been discovered in years when there is no clear indication in ice cores or elsewhere that climatically significant volcanic activity had been occurring. Therefore, although some frost damage may be due to other causes (LaMarche and Hirschboeck 1984: 126), it also may point to gaps in the chronologies of volcanism that have been compiled from other sources (Bradley and Jones 1995b: 617–618). Moreover, because weather patterns over western North America are closely related to those over northeast Asia and the northern Pacific (Lough, Fritts, and Wu 1987), the record of frost damage compiled by LaMarche and Hirschboeck also may help us to understand more about the impact of volcanism on short-term climatic change in East Asia.[8]

The same may be true of other tree-ring evidence from western North America. For example, in a series of articles published since 1987, Louis A. Scuderi has argued that the extremely narrow growth rings sometimes found in another type of subalpine pine *(pinus balfouriana)* in the Sierra Nevada Mountains of California are

> *directly related to the presence of volcanic aerosols in the atmosphere* . . . and represent an independent and extremely well-dated proxy record of climatically effective eruptions and general opacity variations in the northern mid-latitude belt (30° to 50° N) over the past 3000 years *and support the concept of a volcanic mechanism of climate forcing* (Scuderi 1990b: 69–70; see also Scuderi 1987a, 1987b, 1990a, 1992, 1993; Bradley and Jones 1995b: 616–618 [my emphasis]).

It should be noted that this "northern mid-latitude belt" covers all of Korea and Japan and virtually all of China north of a line running from the east coast of the country just north of Ningbo (in modern Zhejiang province) through central China just south of Wuhan and through western China just south of Chengdu. In China, then, all of the Man-

[8] Here it might be noted that the same year during which the "Krakatoa effect" struck subalpine bristlecone pines in the western United States (1884), there was a cool wet summer and a "notably bad harvest" in the key rice-producing area of northeastern Japan (Arakawa 1957b: 212; Takahashi and Nemoto 1978: 184; Kondo 1988: 776). That same year also saw an unusual summer snowfall in central Korea as well as the highest annual precipitation in Seoul for any single year between 1879 and 1897 (Kojŏn Kanhaenghoe 1957, I: 131; Lamb 1977: 633). Precipitation levels in North China in 1884 also appear to have been above average, with floods afflicting several areas along the Yellow River in modern Hebei and Shandong provinces (Wang and Zhao 1981: 279; Satō 1993: 410). Finally, recent research on tree-ring densities from around the world indicates that the summer of 1884 was the thirteenth coldest in the northern hemisphere as a whole in the past 600 years (Briffa et al. 1998: 450).

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churian Plain, the North China Plain, the Huai River Valley, and much of the Sichuan Basin and the Yangzi River Valley lie between 30° and 50° N. Therefore, if Scuderi is correct in his general assumptions, extremely narrow tree rings in *pinus balfouriana* from the Sierra Nevada Range may provide clues to climatic conditions in much of East Asia at the time the rings were produced. Between 1200 and 1699, there appear to have been at least fourteen years in which significant frost damage, extremely narrow tree-ring growth, or both occurred in the mountains of the western United States (Appendix D).

Subsequent research on tree-ring densities in the United States and Canada has allowed scholars to reconstruct "summer half-year [April–September] mean temperatures" in western North America for every year between 1600 and 1982 (Briffa, Jones, and Schweingrüber 1992). Of particular interest here is the fact that this research not only confirms many of the findings by LaMarche, Hirschboeck, and Scuderi concerning the likely connection between volcanism and tree-ring growth in the western United States, but it also indicates that from 1600 (the year this particular study begins) to 1699, volcanic activity may have contributed to exceptionally cool summers in various parts of western North America in 1601, 1612, 1621, 1641, and 1643 (Briffa, Jones, and Schweingrüber 1992: 746–747). Moreover, even more recent research on tree-ring densities from around the world suggests that over the three centuries between 1400 and 1699, volcanic activity contributed to exceptionally cool summers in the northern hemisphere *as a whole* during the following years: 1414, 1438, 1446–48, 1453, 1460–63, 1467–68, 1474, 1495, 1544, 1587, 1589, 1601–02, 1636, 1641–43, 1666–69, 1675, 1686, 1695, and 1698–99 (Briffa et al. 1998: 450; Appendix E).

The "ENSO" Phenomenon and Variations in Solar Radiation

Although the preceding discussion has emphasized the climatic impact of sulfur-rich volcanic activity, several other factors that affect global climate on a short-term basis also should be taken into account. Perhaps the most important of these is the so-called "El Niño–Southern Oscillation" or "ENSO" phenomenon. Briefly stated, the terms "Southern Oscillation" and "ENSO" refer to

> the cycle of waxing and waning in the strength of the pressure gradient across the Pacific, between the Indo-Australian low and the South Pacific high. Usually this pressure gradient leads to convergence of

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> low-level air, and vigorous convection, over the western Pacific. . . . Every few years this circulation is weakened, quickly and substantially, leading to the convection zone moving into the central Pacific and the sea surface temperature (SST) of the tropical Pacific increasing by up to 4° C, particularly in the east. Such an occurrence is called an El Niño. These changes have a major impact on the climate of the whole Pacific Basin and beyond (Bigg 1995: 117).

During the powerful El Niño of 1982–83, there were severe droughts in Australia, Indonesia, the Philippines, India, Sri Lanka, and southern Africa as well as heavy rains in Ecuador, northern Peru, Cuba, and on the West and Gulf Coasts of the United States (Canby 1984: 154–155). Although every El Niño, which also are known as "warm" ENSO events, is different in strength and geographical reach, many of these same climatic anomalies reappeared during the extended El Niño of 1991–94.

The reverse of an El Niño generally is known as a La Niña, a "cold" ENSO event that is characterized by below-normal sea surface temperatures in the eastern Pacific, dry and cool conditions in the northwestern portions of South America, and "anomalously heavy rainfall" in eastern and northern Australia, Indonesia, India, and parts of eastern Africa (Quinn 1992: 127–128). For students of short-term climatic change, it is important to note that "warm" and "cold" ENSO events appear to represent the

> two poles of the [Southern Oscillation] "cycle," with a pronounced tendency for the system to go from one extreme to the other in adjacent years. . . . Thus, one extreme of the [ENSO phenomenon] seems to set the condition in the ocean/atmosphere system for the transition to the opposite extreme in the following year. This is reflected in the fact that warm and cold events are frequently seen to follow each other (Diaz and Kiladis 1992: 11).

These facts are relevant here for several reasons. First, although ENSO events do not appear to have a significant impact on climatic conditions over East Asia *as a whole,* there is some evidence that parts of southeastern China, southern Korea, and southern Japan experience above average rainfall during warm ENSO events and below average rainfall during cold events (Kiladis and Diaz 1989: 1076–1077; Diaz and Kiladis 1992: 14). Perhaps even more important is the fact that a number of notable climatic disasters in East Asian and world history appear to have occurred when global weather patterns were being affected simultaneously by dense veils of volcanic emissions in the atmosphere and by "strong" or "very strong" ENSO events in the Indo-Pacific basin.

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Several periods during which these two phenomena occurred together will be discussed in Part Two below.

Another subject of great interest to students of short-term climatic change has been variations in solar radiation. As one specialist has written, "It is evident that changes in climate over time are likely to be related to variations in ground-level solar intensity, which could be due either to changes in solar output or to changes in the transparency of the atmosphere. Calculations suggest that a decrease of 1 per cent in the radiation output of the sun would be enough to lower global average temperature by 1 or 2° C" (Grove 1988: 365). Beginning with publications by Gustav Spörer and E. W. Maunder in the late nineteenth century, scholars have been aware that at certain times in history there appear to have been significant interruptions of the normal sunspot cycle (Eddy 1997). Among the periods when sunspot activity is thought to have been particularly weak are the so-called Wolf Minimum (c. 1282–1342), the Spörer Minimum (c. 1450–1534), and the Maunder Minimum (c. 1645–1715). This is of particular interest here because some summers during each of these periods are known to have been exceptionally cool (Appendix E). However, as Grove also has pointed out, "while there are striking coincidences in timing [between sunspot activity and fluctuations in global temperatures], they are not perfect" (Grove 1988: 367). For example, although there were a number of very cold summers in the northern hemisphere at the beginning of the Spörer Minimum in the mid-fifteenth century, the early years of the sixteenth century were rather warm (Appendix E). Similarly, although there were six very cold summers in the northern hemisphere during the Maunder Minimum (1666–67, 1669, 1695, 1698–99), all six occurred when dust-veil, ice-core, and tree-ring data suggest there also were heavy veils of volcanic emissions in the atmosphere (Appendixes A, E, F). Moreover, four other very cold summers in the seventeenth century (1601, 1641–43), all of which also coincided with notable volcanic activity, occurred before the date generally accepted for onset of the Maunder Minimum (1645).

Part Two: Case Studies

Introduction

Although there are numerous gaps in, and overlaps among, the various chronologies discussed above, if we combine the years in which 1) there were dense (DVI ≥ 100) volcanic dust veils over the northern hemisphere, 2) strong sulfate acid spikes were deposited in two or more

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northern hemisphere ice cores, 3) notable frost-ring damage or exceptionally narrow tree-ring growth occurred in the mountains of the western United States, 4) tree-ring densities indicate there was an exceptionally cool summer in western North America, and 5) tree-ring densities indicate there was an exceptionally cool summer in the northern hemisphere as a whole, then between 1200 and 1699 there may have been at least 160 years that experienced at least some *short-term* climatic change caused by the presence of volcanic aerosols in the atmosphere (Appendix F). Although an examination of each of those years cannot be attempted here, several key clusters of years will be discussed in the case studies that follow.

C. 1225–1233

Tree-ring research has revealed a significant frost ring event in the western United States during the summer of 1227, thus indicating that one or more climatically effective eruptions may have occurred somewhere in the world between 1225 and the first half of 1227 (LaMarche and Hirschboeck 1984: 122–125). That possibility seems confirmed by recent ice-core research which reveals significant sulfate spikes in two or more northern hemisphere locations in 1228 (± 2 years) (Robock and Free 1996a; Zielinski et al. 1994: 949; Appendix C; Appendix F). Additional northern hemisphere ice-core acid spikes dated to 1230, 1231, and 1232 (all ± 2 years) indicate that volcanic activity of some significance probably occurred between 1228 and late 1233 as well. Among the volcanoes known to have been active during this period are: 1) Iceland's Mt. Reykjanes, which began a lengthy and "severe" (VEI 3?) eruption during the summer of 1226; 2) Mt. Zao on the Japanese island of Honshū, which first erupted in the autumn of 1227 (VEI 3) and then again in the autumn of 1230 (VEI 2?); 3) Mt. Aso on the Japanese island of Kyūshū, which erupted in 1229 (VEI 2); and 4) Iceland's Mt. Reykjaneshryggur, which erupted in 1231 (VEI 3) (Simkin and Siebert 1994: 188). It also may be significant that tree-ring data from northern Scandinavia and ice-core data from the Qinghai-Tibetan Plateau of northwestern China both indicate a sharp drop in summer temperatures in those regions during the late 1220s and early 1230s (Briffa et al. 1990: 437; Thompson et al. 1993: 147). In later periods, such declines in temperature have often been associated with the presence of volcanic emissions in the atmosphere. Finally, there is also some evidence that a strong warm ENSO event occurred in the Indo-Pacific Basin in 1230 (Quinn 1992), an event that may have caused

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increased rainfall in southeastern China, southern Korea, and southern Japan. Moreover, given what is now known about the close connections between warm and cold ENSO events, very dry conditions may have existed in some of those regions a year or so later.

As students of medieval Japan will be aware, climatic and economic conditions there during the late 1220s and early 1230s often were terrible. In 1226, for example, the Kyōto region suffered from high winds, heavy rains, and then from an extended drought. This was followed by "incessant" rains and floods in the spring of 1227 and again in the summer of 1228. Then, in the autumn of 1229, inadequate rainfall is said to have led to serious food shortages in some regions. Sharp fluctuations in temperature and precipitation in 1230 were followed by heavy rains and floods in 1231, all of which helped to produce what many scholars regard as one of the most serious famines in Japanese history (Arakawa 1955b; Nakajima 1976: 7; Arakawa and Usami 1985: 319; Sansom 1958: 393).[9] Climatic and economic conditions in neighboring Korea were often quite bad during this period as well. For example, an unusual cold spell on the peninsula in the summer of 1226 was followed by famine in 1227–28. By the early months of 1230, a year in which food shortages in Japan were reaching crisis proportions, a severe famine had begun in Korea as well (Kojŏn Kanhaenghoe 1957, I: 137, 147).

In northern China, the Jin dynasty (1127–1234) government, which already was under heavy military pressure from the Mongols, was forced to deal with unusually cold temperatures early in 1226 and then, as the year progressed, with locust infestations, drought conditions, and, finally, with heavy rain and hail. Unusual cold and rain also were a problem in southern China in 1226, a year that reportedly brought exceptionally murky skies, high winds, and floods to parts of Zhejiang and Anhui. High winds and a late frost during the summer of 1227 are said to have sharply reduced grain production in northern China.[10] The early months of 1228 again were cold in the north, and during the late spring or early summer of that year heavy rain and hail severely

[9] Nakajima ascribes the great Japanese famine of 1230–31 to *dai reigai* or "severe [crop] damage from cold" (Nakajima 1976: 7). As Arakawa, Kondo, Lamb, and others have pointed out, in more recent times such crop damage in Japan often occurs during summers just after great volcanic eruptions when changes in atmospheric circulation cause "anomalous" northeasterly winds to develop over the northwestern Pacific (Arakawa 1957b; Kondo 1988; Murata 1995).

[10] Unusually cool temperatures during the summer of 1227 also brought "frost damage" to subalpine pine trees in the mountains of California (LaMarche and Hirschboeck 1984: 124).

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damaged trees and agricultural production in parts of Henan. Heavy snows in northern China during the early months of 1232 were followed by unusual cold during the fifth lunar month, a time that is said to have felt "like winter" (Tuo-tuo et al. 1975, II: 544–545; Satō 1994: 163–164).

Although few, if any, records of climatic conditions in areas north of the so-called Great Wall Frontier exist for this time, the fact that both Korea and China experienced unusual rain, snow, and cold in 1226–28 and 1232, and the fact that the great Japanese famine of 1230–31 appears to have been caused, at least in part, by exceptionally cool summer temperatures, indicate that Northeast Asia as a whole may have been experiencing significant changes in atmospheric circulation that not only altered wind and precipitation patterns but also allowed Arctic air to penetrate further south than normally would have been the case. That possibility seems borne out by what is known about conditions in other parts of Eurasia at this time. For example, there were serious floods and food shortages in northwestern Russia in 1228 while other regions of the country are known to have experienced floods, sharp fluctuations in temperature, and severe famine in 1229–30 (Borisenkov and Pasetskii 1983: 134–135). The *Nikonian Chronicle* describes the situation in Novgorad during these years:

> And God's wrath arose, and the land was devastated: there was rain day and night, from Ascension Day to St. Elias' Day [1230]; and the cold came, and there were great freezes which destroyed all the rye. . . . And there was a famine, and people died from the great starvation so that it was even impossible to bury them. . . . The starvation spread not only to the land of Novgorad but to all the Russian land, with the exception only of Kiev. The wrath of God was such that the people not only ate the dead but even the living would kill and eat each other; and they ate horse flesh, dogs, cats, and other animals. Whatever one could find, he would eat—moss, pine, elm, linden bark and leaves. Evil people, learning that someone had grain, would force their way into that place, pillage, and kill; and divine wrath spread. People died all over the whole land, and there was no numbering them; and thus it continued for two years [1229–30] (Zenkovsky 1982–89, II: 303–304; see also Vernadsky 1948: 315).

Although economic conditions in western Europe do not appear to have been as desperate as those in Russia or Japan, there were a series of very poor harvests in England, the Low Countries, France, Lorraine, Bavaria, and the Rhine Valley during the mid- and late 1220s (Parry 1978: 137; Farmer 1988: 722; Abel 1980: 18). Across the Mediterra-

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nean the failure of the summer monsoon over Ethiopia in 1230 caused extremely low water levels and, in all probability, severe agricultural problems in the Nile River basin and other parts of northeastern Africa (Quinn 1992: 141; Herring 1979: 79).[11] Further afield, tree-ring evidence from the North Platte region of North America indicates that there was an extended period of drought there during the 1220s and early 1230s, a situation reminiscent of conditions in that region a few years earlier when changes in wind and precipitation patterns are thought to have caused farming peoples in the northern Great Plains to migrate to more favorable locations further south (Wedel 1986: 44–46).

Although it is risky to attempt to demonstrate a clear connection between climatic conditions and economic and political changes in a poorly documented period nearly eight centuries in the past, it nevertheless is the case that the years from the mid-1220s to the early 1230s were eventful ones in Eurasian history. One of the most notable stories at the time concerned the Mongols. Despite the death of Chinggis Khan in 1227, over the next seven years the Mongol military completed the conquest of the Xi Xia state in northwestern China, secured control over much of modern Manchuria, conducted a series of campaigns in Korea, destroyed the remnants of the Jin dynasty in north China, and carried out exploratory invasions of southern Russia and Iran (Allsen 1994: 364–372). Of interest here is the fact that virtually everywhere the Mongols attacked during this brief period recently had experienced, or was just then experiencing, significant economic problems caused, at least in part, by "anomalous" climatic conditions. The extent to which the Mongol homeland itself was being affected by those same anomalies remains to be studied. Given what is now known about conditions in other parts of the northern hemisphere at that time, however, Gareth Jenkins's comment that the Mongols's "enthusiasm for the task of conquest may well have been fueled by a climatic defeat at their backs" (Jenkins 1974: 226) may be as relevant for this period as for the late twelfth and early thirteenth centuries that were his primary focus.

[11] While doing research for this study, I noted a number of occasions (c. 1180, 1200, 1230, 1332, 1403, 1450, 1641, 1782, and 1832, for example) when climatic conditions in the Nile River Valley and Japan appear to have deteriorated sharply at virtually the same time. In response to a question concerning these coincidences, Syukuro Manabe explained that during the late spring and early summer, "the easterly jet stream in the upper troposphere extends from Japan to Egypt and may affect the climate of both regions in a somewhat similar fashion" (Manabe 1997; see also Yamamoto 1978: 61).

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C. 1256–1262

> In recent years, there are unusual disturbances in the heavens, strange occurrences on earth, famine and pestilence, all affecting every corner of the empire and spreading throughout the land. Oxen and horses lie dead in the streets, the bones of the stricken crowd the highways. Over half the population has already been carried off by death, and in every family someone grieves. . . . [Despite prayers and relief projects], famine and disease rage more fiercely than ever, beggars are everywhere in sight, and scenes of death fill our eyes. Cadavers pile up in mounds like observation platforms, dead bodies lie side by side like planks on a bridge (Nichiren, *Risshō ankoku ron*, quoted in Yampolsky 1990: 13–14).

The above passage was written by the Japanese Buddhist leader Nichiren in an essay he began drafting in 1257 and eventually submitted to an influential figure in the shogunal capital of Kamakura during the summer of 1260. What Nichiren is describing here are the tragic results of a series of "natural" and other disasters that had struck Japan over the previous few years: 1) high winds, torrential rains, floods, and destructive landslides in the summer of 1256; 2) an outbreak of dysentery that same year that killed the shogun and many others in Kamakura; 3) a string of powerful earthquakes in east-central Japan in 1257; 4) drought conditions in many parts of the country during the summer of 1257; 5) destructive wind storms, floods, and unusually cool weather that led to poor harvests in some areas in 1258; 6) violent wind and rain storms that severely damaged crops in the summer of 1259; and 7) a nationwide famine in 1259–60 that appears to have been caused by unusually cool summer temperatures (Yampolsky 1990: 11; Jannetta 1987: 117, 148; Arakawa and Usami 1985: 320; Nakajima 1976: 7; Sansom 1958: 417).

At virtually the same time that Nichiren was drafting his famous essay, the Benedictine monk Matthew Paris was compiling his "Great Chronicle" *(Chronica Majora)* in the abbey at St. Albans just north of London. Like Nichiren, Paris paid close attention to contemporary climatic and economic conditions:

> [1256] was beyond measure stormy and rainy. . . . From [mid-August 1256] to [early February 1257] the rain ceased not to fall daily in deluges, which rendered the . . . fields barren. . . .
>
> [1257] throughout was barren and meagre; for whatever had been sown in winter, had budded in spring, and grown ripe in summer, was

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> stifled and destroyed by the autumnal inundations. . . .[12] The land lay uncultivated, and great numbers of people died from starvation. . . . [During the autumn and winter] . . . uninterrupted heavy falls of rain and mist obscured the sky.
>
> [From late January] till the end of March [1258], the north wind blew without intermission, a continual frost prevailed, accompanied by snow and such unendurable cold, that it bound up the face of the earth, afflicted the poor, suspended all cultivation, and killed the young of the cattle to such an extent that it seemed as if a general plague was raging amongst the sheep and lambs. . . . [T]he north wind blew incessantly [during the spring and early summer]. . . . Owing to the shortage of food, a very large number of poor people died; and dead bodies were found everywhere. . . . [A]t the end of July and the beginning of August . . . such misery, want, and famine prevailed, that those who usually aided others were amongst the unfortunates who perished from want. . . . This year throughout was very unlike all previous ones, bringing disease and death, stormy and rainy; to such an extent that although in the summer a fair promise of abundant crops . . . was given, in the autumn the continual heavy rains spoiled the corn, fruit, and all kinds of pulse once more; and in Advent, in some parts of England . . . , the barns were empty and the crops remained standing for cutting, but entirely spoiled: for as the grain germinated, the ear and the straw rotted together, and as men died from the want of corn, so the cattle died from want of fodder. . . . (Matthew Paris quoted in Rothwell 1975: 110, 119, 131, 141, 822–823; see also Mollat 1986: 158).

Paris's frequent references here to "unusual" summer and autumn storms and to the fact that winds blew "incessantly" from the north suggests that during much of the late 1250s, southern England and other parts of western Europe were experiencing significant "anomalies" in wind and precipitation patterns. The heavy rains and cool summer temperatures that affected Japan during this period suggest the same also was true of northeast Asia. Reports from Korea provide further evidence of this. For example, heavy rains on the peninsula during the summer of 1256 were a factor in causing severe food shortages that lasted well into 1257. Then, after a brief respite, incessant

[12] According to Wilhelm Abel, "In 1257, when the highest prices of the century were being paid in England, similar famine prices reigned in France, Swabia, and the Rhineland" (Abel 1980: 18). For a brief discussion of the "food shortages, high prices, and riots" that occurred in several Italian cities during this period, see Mollatt 1986: 158.

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rains during the summer of 1258 reduced that year's harvest to the point that famine and even cannibalism were reported in the capital region early in 1259. Economic difficulties continued during the summer of 1260 when severe drought and infestations of locusts once again reduced agricultural production on the peninsula (Kojŏn Kanhaenghoe 1957, 1: 147).

Perhaps because of widespread Mongol military operations, reports on climatic and economic conditions in China are not as plentiful or detailed as those for many other periods. However, the summer of 1257 and the early months of 1258 appear to have been very dry in southeastern and south-central China, especially in parts of Jiangsu, Anhui, and Hubei.[13] In 1259, there were extended rains in northern China during the fourth lunar month, severe flooding in the south during the fifth lunar month, and, at some point during the year, significant flooding in Anhui and Zhejiang. Heavy rains in the Yangzi delta during the fourth lunar month of 1260 quickly gave way to such dry conditions that Southern Song (1127–1279) authorities soon were sponsoring special ceremonies to pray for rain. Later that same year food shortages in parts of Shandong led authorities there to undertake extensive relief operations (Satō 1994: 166–167).

As had been true of the crisis years around 1230, tree-ring data from Scandinavia and ice-core data from the Qinghai-Tibetan Plateau suggest that summer temperatures in those widely scattered regions were very cool during the late 1250s and early 1260s (Briffa et al. 1990: 437; Thompson et al. 1993: 146), a phenomenon that probably was associated with volcanic emissions in the atmosphere. Not only do Korean meteorological observations point to the existence of such emissions in 1256, 1258, and 1260 (Kojŏn Kanhaenghoe 1957, I: 96), but ice-core research has revealed that one or more powerful eruptions probably occurred somewhere in the world between late 1256 and 1259. Moreover, those eruptions left extraordinarily strong sulfate acid spikes in multiple ice cores for each year between 1258 and 1262 (± 2 years) (Palais, Germani, and Zielinski 1992; Robock and Free 1996a; Appendix C). Analysis of volcanic glass shards found in ice cores from Antarctica and Greenland suggests that one source for these acid spikes may have been a powerful but not yet firmly dated eruption at Mt. El Chichón in south-central Mexico (Palais, Germani, and Zielinski

[13] The summer of 1257 also was very dry in much of Japan (Arakawa and Usami 1985: 330).

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1992).[14] In addition to this ice-core evidence, exceptionally narrow growth rings found in *pinus balfouriana* from the Sierra Nevada Mountains in California and dated to 1257 indicate that another climatically effective eruption or series of eruptions may have occurred somewhere in the world between late 1255 and the first half of 1257 (Scuderi 1990b: 73). In this connection, it is known that a "severe" (VEI 3) eruption occurred on the Arabian peninsula in the summer of 1256. To what extent this eruption affected climatic conditions in the Mid-

[14] The likelihood that a powerful, "sulfur-rich" eruption occurred at El Chichón during the late 1250s is intriguing because of the very large (VEI 5), "sulfur-rich" eruption that took place there in the spring of 1982. The 1982 eruption produced what was called, at least until the eruption of Mt. Pinatubo in the Philippine Islands in June 1991 (VEI 6), the "Dust Cloud of the Century" (Robock 1983). What is particularly important about El Chichón in 1982 is that both the eruptive sequence and the "dust cloud" it produced were closely monitored by ground stations, by specially equipped aircraft and balloons, and by orbiting satellites (Robock 1983; Matson and Robock 1984; Rampino and Self 1984; Hayashida, Kobayashi, and Iwasaki 1984; Hirano et al. 1984; McCracken and Luther 1984; and McPeters, Heath, and Schlesinger 1984). Using weather satellites, Matson and Robock tracked the visible image of the dust cloud as it moved westward and completely circled the globe in twenty-one days. Satellite photography indicates that the leading edge of the dust cloud reached Taiwan and parts of southern China by 17 April 1982. By 24 and 25 April, the main cloud had moved far enough to the north to cover a substantial portion of central China (Matson and Robock 1984: 124–125; Rampino and Self 1984: 54). To the north and east, ground stations in southern and central Japan (at Fukuoka and Nagoya, respectively) detected "enormous increases" in stratospheric aerosols during the second half of 1982 and in early 1983 (Hayashida, Kobayashi, and Iwasaki 1984; Hirano et al. 1984). Given the latitudes of Fukuoka and Nagoya, this probably means that much of central and northern China and all of Korea experienced similar increases in aerosol readings at this time. In its initial circuit of the earth, the main cloud from El Chichón also passed over much of mainland Southeast Asia, the Bay of Bengal, northern India, the Arabian peninsula, the highlands of Ethiopia, and the Sahel region of West Africa before crossing the Atlantic Ocean and reaching Mexico from the east (Rampino and Self 1984: 54).

Analysis of the climatic effects of this eruption is complicated by an earlier (December 1981) "sulfur-rich" eruption (VEI 3) at Mt. Nyamuragira in central Africa (Krueger, Schnetzler, and Walter 1996), by later eruptions at Indonesia's Mt. Galunggung (May 1982, VEI 4), Mt. Raung (July 1982, VEI 3), and Mt. Soputan (August 1982, VEI 3), and by the subsequent development of a powerful warm ENSO event in the Indo-Pacific Basin. Nevertheless, one recent study suggests that even when the global warming caused by the 1982–83 ENSO event is taken into account, the "dust cloud" from El Chichón probably caused "substantial posteruption cooling" in the northern hemisphere (Mass and Portman 1989: 583–585). Moreover, as was the case following the great eruption or eruptions of the late 1250s and early 1260s, the years following the 1982 eruption at El Chichón found global weather patterns greatly disturbed. For example, there were heavy rains and floods in southern China (May 1982); southern Japan (July–September 1982); western Europe (October–December 1982); northern Europe (January–February 1983); in Denmark, the Low Countries, Germany, and France (March–May 1983); central China (May–July 1983); and South Korea (September 1984). There also were unusually cool summer temperatures in southern Japan (1982), central China (1983), and northern Japan (1982–84) (LeComte 1983; LeComte 1994; LeComte 1995; and Wagner 1983: 6–13).

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dle East is not clear, but unusually heavy rains that year contributed to severe flooding in the Tigris River Valley (Ashtor 1976: 245). Other volcanoes thought to have been active during the 1250s and 1260s include Mt. Pico de Orizaba in Mexico and Mt. Pelée on the island of Martinique. There also was a VEI 3 eruption at Iceland's Mt. Katla in 1262 (Simkin and Siebert 1994: 188).

As was the case during the late 1220s and early 1230s, the political history of Eurasia during the 1250s and 1260s was a tumultuous one. In Japan, relief operations by the Kamakura shogunate were complicated by widespread banditry, peasant unrest, and violent uprisings led by militant Buddhist clergy (Kawasaki et al. 1990: 346–348). In Korea, the Mongol invasions of 1254 and 1256 were so destructive that they helped to keep much of the peninsula in turmoil for the next several decades (Eckert et al. 1990: 92–93). Despite their military successes in Korea and elsewhere, the Mongols experienced problems of their own during this period. Among the more notable of those was the conflict between Khubilai Khan, who was in command of Mongol forces in northern and western China, and his brother Arigh-böke, who controlled the region in and around modern Mongolia. This struggle between them lasted from 1260 to 1264, and when it was over Khubilai had emerged victorious, in part, it is said, because he had been able to deny his brother access to the resources of China (Rossabi 1994: 423–424; Morgan 1986: 117). However, another element in Arigh-böke's defeat may have been the fact that temperature levels in northern and central portions of Eurasia apparently were declining sharply in the late 1250s and early 1260s (Briffa et al. 1990: 447; Thompson 1991: 214). If so, this probably made it much more difficult for forces based in Mongolia to acquire the food, fodder, and other supplies necessary to mount and sustain a military offensive.[15]

C. 1444–1465

Recent analysis of northern hemisphere ice cores has revealed significant sulfate acid spikes in at least two locations during the following years: 1445, 1453–54, 1456–57, and 1459–64 (all ± 2 years). A similar analysis of southern hemisphere ice cores has found notable acid spikes in 1444, 1451–53, 1461–62, and 1465 (all ± 2 years) (Robock and Free

[15] Here it also might be noted that Mongol forces under Hülegü sacked Baghdad and murdered the last Abbasid Caliph in 1258, just two years after serious flooding had severely damaged agricultural production in the Tigris Valley (Ashtor 1976: 245).

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1996a, 1996b; Appendix C). These findings take on new significance when placed alongside tree-ring data from the same period. Not only do LaMarche and Hirschboeck find clear evidence of major frost damage in bristlecone pines from both the Great Basin and the Rocky Mountains of the western United States in 1453, but even more recent research on tree-ring densities from around the world suggests that there were no fewer than twelve unusually cool summers in the northern hemisphere between 1444 and 1463 (Appendix E). Especially notable among those were 1446, 1448, and 1453, which one group of researchers has ranked as the thirtieth, nineteenth, and fourth coldest summers respectively in the northern hemisphere as a whole in the past 600 years (Briffa et al. 1998: 450). This cooling trend also appears to have been accompanied by sharp fluctuations in wind and precipitation patterns in many parts of the world. For example, on the mid-Atlantic coast of North America, a notable dry spell between 1451 and 1456 immediately was followed by rather wet conditions between 1457 and 1461 (Stahle et al. 1998). During the 1450s there also was a savage drought and famine in the Mexican highlands (Gill 1994)[16] as well as a period of "very dry" conditions in the North Platte region of western Nebraska (Wedel 1986: 45).

Although there can be little doubt that many of the ice-core acid spikes and tree-ring anomalies discussed above were caused by climatically effective eruptions, the precise location and timing of some of those eruptions still is unclear. However, there was a "cataclysmic" (VEI 4) eruption in Japan in 1442, another VEI 4 eruption in the Azores and a "severe" (VEI 3) eruption there in 1444, three VEI 3 eruptions at Mt. Kelut on the island of Java in 1450, 1451, and 1462, and one eruption of unknown magnitude in the northern Philippines in 1464. In addition, uncorrected carbon-14 dating indicates that a "paroxysmal" (VEI 5?) eruption probably occurred at Alaska's Mt. Aniakchak sometime around 1450. Recent research also has dated a "colossal" (VEI 6) volcanic event at Mt. Kuwae in the southwestern Pacific to 1452 (± 10 years) and a VEI 3 eruption at Peru's El Misti to 1454 (± 16 years) (Simkin and Siebert 1994: 190; Robin, Monzier, and Eissen 1994; Monzier, Robin, and Eissen 1994). Complicating the study of global climatic conditions during this period is the fact that the year 1450 not only generally is accepted as the beginning of the Spörer Sunspot Minimum but also a year that appears to have witnessed a very strong warm ENSO event in the Indo-Pacific basin (Quinn 1992).

[16] I am grateful to Yoav Me-Bar for this reference.

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Not surprisingly, there is strong evidence that climatic conditions throughout the northern hemisphere (and probably the southern hemisphere as well) were greatly disturbed between the mid-1440s and early 1460s.[17] In much of northern and western Europe, for example, the 1450s were marked by at least four unusually cool and wet summers. Indeed, when decadal averages are compared, summers in both England and Germany during the 1450s rank among the wettest recorded there between 1100 and 1969 (Lamb 1977: 562). The economic significance of this is that in western and central Europe, poor harvests usually are due not to drought but rather to "prolonged cold spells in spring and to cool and wet summers, especially during the harvest season, when such conditions may cause the grains to rot in the fields" (Flohn 1981: 312; see also Ladurie 1971: 288–292; Appendix E). During this same period, Europe also experienced several extremely cold winters, with those of 1458 and 1460 regarded as being particularly severe. Especially hard hit by the cold were communities in Finland, Sweden, Denmark, Norway, Iceland, and Greenland, although parts of England, Scotland, the Low Countries, and Germany also suffered (Lamb 1982: 191; Lamb 1977: 454–460, 562, 566–568; McGovern 1981: 421). Further to the east, there were several very cold winters in Russia during the 1440s, a severe drought in the Moscow region in 1451, and famine conditions in Novgorad in 1445–46, 1453, and 1462, in Smolensk in 1447, in Moscow in 1451 and 1454, and in all of Russia in 1448 (Borisenkov and Pasetskii 1983: 165–168). To the south in the Balkans, the bitterly cold winter of 1443–44 forced the Hungarian army to halt its campaign against the Ottoman Turks and retreat back across Serbia (Heywood 2000).

Nor did southwestern Europe escape from this period unscathed. In Spain, where there apparently were extraordinarily sharp fluctuations in precipitation between the mid-1440s and the 1470s (Martín-Vide and Barriendos-Vallvé 1995: 217), crops were damaged by severe storms in 1446 and by drought in 1451–52. Over the following decade heavy rains, floods, locust attacks, and other disasters led to a series of food riots, perhaps the most famous of which occurred in Seville in the

[17] The climatic problems of the 1440s, 1450s, and 1460s had been preceded during the 1430s by significant changes in global weather patterns that had led to poor harvests, outbreaks of epidemic disease, and demographic contraction from literally one end of Eurasia to the other. I have discussed these developments briefly in Atwell 1993. See also Twitchett and Grimm 1988: 310; Satō 1993: 242–249; Mackay 1981: 359–367; Hatcher 1977: 55–57; Lamb 1977: 457–459; Lamb 1982: 187–196; Dols 1977: 143–235; and Nagahara 1974: 210–235.

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spring of 1462 (Mackay 1981: 368). Economic conditions in southern France also were grim during this period and apparently would remain so almost to the end of the century (Ladurie 1974: 14). Across the Mediterranean very low water levels in the Nile Valley between 1448 and 1452, a situation reminiscent of the crisis years around 1180, 1200, and 1230, undoubtedly contributed to the extraordinarily high food prices that were recorded in Egypt during the early 1450s (Herring 1979: 55; Quinn 1992: 141; Shoshan 1993: 64). Commenting on the economic and political unrest in Cairo during these years, one specialist has written that there "were many great droughts, prolonged periods of low water in the Nile, locusts, famines, and plagues. . . . [I]t was impossible for [Egypt] to recover from these catastrophes" (Hrbek 1977: 61; see also Ashtor 1976: 322–331). The mid-fifteenth century also saw serious economic and political unrest in Iraq (Ashtor 1976: 270–271) as well as the final collapse of the Byzantine Empire in the Balkans and Anatolia. When the Ottomans completed their conquest of Constantinople in 1453, the population of that once great city had dropped to no more than 50,000 from a peak of perhaps 350,000 a few centuries earlier (Chandler 1987: 239; Inalcik 1994: 18).

Students of East Asian history perhaps will have noted some parallels with the above developments. In China, there were "massive famines" in the northwest in 1444–45, famines in the central Yangzi region in 1444, 1446, and 1448, serious flooding in the Huai River Valley in 1444, in the southeast in 1446–47, and on the eastern reaches of the Yellow River in 1448, and severe drought in Zhejiang in 1445 and in the southeast and northwest in 1448 (Twitchett and Grimm 1988: 310–311). Nor did the situation improve much, if any, during the 1450s:

> In 1450 there was famine in Shantung. In 1452 heavy rain and serious flooding affected the Yellow River and Huai River Valleys in Honan, the Northern and Southern Metropolitan regions, and Shantung, and special measures had to be taken to give relief in Chiang-pei. The wet weather continued into 1453 and 1454 and was made worse by an exceptionally cold winter in 1453–54, with very heavy snow in Shantung, Honan, the Southern Metropolitan Region, and Chekiang. The sea off the mouth of the Huai was icebound, and the Huai Valley was badly affected, tens of thousands dying of cold in the area. In Kiangnan at the beginning of 1454, a heavy snowfall lay for more than forty days, and countless people died of cold and starvation in Soochow and Hangchow. Even as far south as Heng-chou in southern Hunan there was continuous snow, which killed off many draft cattle. In 1455 there was widespread drought affecting the Southern Metropolitan Region,

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> Hu-kuang, Kiangsi, Honan, Shantung, Shansi, and Shensi. The next year, 1456, was another abnormally wet year, with continual rain through summer and autumn, which caused the loss of crops and immense damage in the Northern Metropolitan Region, Shantung, Honan, and the Southern Metropolitan Region, while the Yangtze Valley, Chekiang, and Kiangsi suffered serious droughts (Twitchett and Grimm 1988: 336–337; see also Robinson 1999: 93–97).

China's problems persisted over the next few years with famine conditions in Hebei and Shandong in 1457, in Huguang in 1458–59, in Huguang and Guangxi in 1460, in Shaanxi in 1462, in Henan in 1463, and in Hebei, the lower Yangzi Valley, Zhejiang, and Henan in 1465 (Satō 1993: 247–251; Song et al. 1992: 175). Adverse weather conditions and food shortages during these years apparently also caused serious difficulties in Mongol-controlled territory north of the Great Wall frontier (Robinson 1999: 95–96).

Although not as complete as those for China, Korean sources report drought in 1450 and 1452 and famine conditions in 1445, 1452, 1454, and 1458 (Kojŏn Kanhaenghoe 1957, I: 148–149). The situation in nearby Japan may have been even worse. As Nagahara Keiji has put it:

> During a period of abnormal climatic conditions between about 1440 and 1460, parts of Japan were struck repeatedly by [unusual] cold, famine, and outbreaks of epidemic disease. Moreover, it was precisely during these years that the number and intensity of peasant revolts reached their peak. . . . It cannot be denied that the fundamental causes of these uprisings were poor [and sometimes even] disastrous harvests (Nagahara 1988: 314–316).

To provide further details, there was major flooding in Japan in 1448, 1451, and 1460–61, severe drought in some areas in 1447, 1451, 1457, 1459, 1461, and 1465, and an exceptionally cold winter in much of the country in 1458–59. As was the case in many other parts of the northern hemisphere at this time, the ensuing food shortages and loss of life were appalling. For example, in one two-month period during the winter of 1460–61, more than 80,000 people reportedly died from hunger and disease in the city of Kyōto alone (Nagahara 1974: 210–235; Nagahara 1988: 314–316; Arakawa 1955b: 145; Arakawa 1967: 4; Hall 1990: 228).

The middle decades of the fifteenth century were a time of significant political and military upheaval in the northern hemisphere. In

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western Eurasia, for example, one might point to Anglo-French hostilities during the final phases of the Hundred Years War, the outbreak of "civil wars" in England, France, and the Iberian Peninsula, the conquest of Delhi by Afghan invaders in 1451, the Balkan wars of the 1440s, the fall of Constantinople in 1453, and the military campaigns of Ivan III in Russia. In East Asia, there were major peasant uprisings in Japan in 1441, 1447, 1452, 1454, 1456–57, 1459–60, 1462–63, and 1465–66. These were followed in 1467 by the outbreak of the Ōnin War, a protracted struggle (1467–77) that further disrupted economic activity in south-central Japan and set the stage for the "Warring States" period in Japanese history. South of the main Japanese islands in the Ryūkyū Kingdom, there also was considerable economic and political strife during the middle decades of the century (Takara 1993: 55–57). In Korea, the death of King Sejong in 1450 ushered in a series of rapid changes at the Yi dynasty court that included several rounds of political assassination both before and after the usurpation of the throne by King Sejo in 1455. Meanwhile, deteriorating economic conditions in the Korean countryside contributed to a series of "peasant revolts" during the 1460s (Eckert et al. 1990: 108).

In China, anti-government uprisings that began in the border areas of Zhejiang, Fujian, and Jiangxi provinces during the mid- 1440s were not brought under control until the early 1450s. That same period also saw Ming armies mount a series of costly campaigns in southwestern China and northern Burma. In July 1449, after tensions on the northern and northwestern frontiers had been escalating for some time, due in part it appears, to deteriorating climatic conditions, the Oirat leader Esen began a large-scale invasion of China. This led to a disastrous campaign during which the Ming emperor was captured by the invaders and Esen's forces advanced to the walls of Beijing. Although the Ming government managed to establish relatively peaceful relations with the Oirats over the next few years, the dynasty's security problems continued. Between 1450 and 1456, Ming forces were forced to put down anti-government uprisings in Guizhou, Huguang, Guangdong, Fujian, Zhejiang, and Sichuan (Twitchett and Grimm 1988: 336). Moreover, as Robinson recently has pointed out, "bad weather" along the Great Wall Frontier during the late 1450s and early 1460s appears to have caused serious problems for the Chinese and the Mongols living in the area, increasing tensions between the two groups and contributing to Mongol participation in the "abortive coup" of 1461 in which disaffected members of the Ming government attempted to remove the emperor from power (Robinson 1999: 94–97).

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C. 1584–1610

In marked contrast to the mid- and late fifteenth century, the early decades of the sixteenth century appear to have been a time of low volcanic activity (Appendixes A, B, C, D, F). Recent studies of tree-ring densities from around the world suggest that one result of this may have been summer temperatures in the northern hemisphere as a whole that were both warmer and more stable than those that had prevailed during much of the fifteenth century. However, as can be seen in Appendix E, those favorable conditions did not last indefinitely. Indeed, tree-ring studies suggest that there were significant declines in global temperatures during the late 1530s and early 1540s and then again during the late 1560s and early 1570s. Of particular interest here is the fact that during these two periods, both of which appear to have coincided with increased volcanic activity,[18] significant fluctuations in temperature and precipitation as well as serious food shortages occurred in parts of China, Korea, Japan, Mongolia, Russia, the Mediterranean basin, and, based on tree-ring data, in the mid-Atlantic and North Platte regions of North America (Satō 1993: 266–271; Song 1992: 176, 185, 202, 296, 303, 311, 323; Brook 1998: 104–106; Kojŏn Kanhaenghoe 1957, I: 148–149; Arakawa and Usami 1985: 323–324; Borisenkov and Pasetskii 1983: 185–196; Braudel 1972, I: 593; Martín-Vide and Barriendos-Vallvé 1995: 214–217; Stahle et al. 1998; Wedel 1986: 45).

As serious as some of those famines were, however, they pale in comparison to the disasters that struck many parts of the world between about 1584 and 1610. Over that twenty-seven-year period, sulfate acid spikes were deposited in two or more northern hemisphere locations in fourteen different years (all ± 2 years)—1585–86, 1588–90, 1596–97, 1600–02, and 1605–08—and in two or more southern hemisphere locations in three different years—1589, 1597, and 1602 (Appendix C; Robock and Free 1996a). In addition, tree-ring densities from around the world indicate that summer temperatures in the

[18] Although ice-core research has not yet confirmed the incidence of sulfur-rich volcanic activity during the late 1530s and early 1540s, other sources indicate that no fewer than four "severe" (VEI 3) eruptions occurred in various parts of the world between 1535 and 1542 (Simkin and Siebert 1994: 190–191). The situation for the late 1560s and early 1570s seems clearer as strong sulfate acid spikes were deposited in two or more northern hemisphere ice cores in 1568, 1572, and 1575 (all ± 2 years) (Appendix C). This suggests that climatically effective volcanic activity probably occurred somewhere in the world between 1566 and 1576. Of the more than forty confirmed eruptions that occurred during this eleven-year period, no fewer than eight were rated as "severe" (VEI 3) (Simkin and Siebert 1994: 191).

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northern hemisphere fluctuated quite sharply during this period. Indeed, 1601 may have been the single coldest summer in the northern hemisphere in the past 600 years;[19] yet another summer during this period, that of 1587, has been ranked as the seventeenth coldest summer in the northern hemisphere over the same six centuries (Briffa et al. 1998: 450; Appendix E).[20]

As the available ice-core, tree-ring, and temperature data indicate, there was a significant increase in both the frequency and intensity of volcanic activity during the late sixteenth and early seventeenth centuries. For example, between 1582 and 1611, no fewer than twenty-six eruptions have been given VEI ratings of 3 ("severe") or higher. Among the more notable of these eruptions were: 1) the "cataclysmic" (VEI 4) events at Mt. Fuego in Guatemala in 1582, at the Colima Volcanic Complex in Mexico in 1585 and 1606, at Mt. Ruiz in Colombia in 1595, and at Mt. Momotombo in Nicaragua in 1605; 2) the "paroxysmal" (VEI 5) eruptions at Indonesia's Mt. Kelut in 1586 and Mt. Raung in 1593; and 3) the "colossal" (VEI 6) eruption at Peru's Mt. Huaynaputina in 1601. In addition, uncorrected carbon-14 data indicate that yet another VEI 6 eruption probably occurred in the Southwest Pacific at some point between 1560 and 1600 (Simkin and Siebert 1994: 192–193). Once again complicating the climatic history of the period is the fact that "moderate" (M) or "strong" (S) warm ENSO events appear to have occurred in the Indo-Pacific basin in 1585 (M), 1589–91 (S), 1596 (M), 1600–01 (S), 1604 (S), and 1607–08 (S) (Quinn 1992: 123). As was noted earlier, some of these events probably were followed in subsequent years by significant cold ENSO events.

The late sixteenth and early seventeenth centuries brought severe climatic and economic problems to many parts of Europe. Not only did much of the continent experience unusually cool temperatures, poor harvests, and increased storminess during the 1580s, the last of which included the powerful gales that apparently doomed the Spanish Armada during the summer of 1588 (Lamb 1977: 466; Lamb 1982: 208–209), but, as David Hackett Fischer has noted:

> In 1591, the weather turned wet and cold. European peasants watched helplessly as their wheat and rye were beaten down in the fields, and their hay crops rotted in the meadows. The same thing happened the

[19] Significant frost-ring damage dated to 1601 has been discovered in a wide variety of locations in the western United States (Lamarche and Hirschboeck 1984: 122–125).

[20] Below normal tree growth has been detected in California's Sierra Nevada Mountains for all but one year (1602) between 1597 and 1610 (Scuderi 1990b: 73).

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> next year, and the year after that, and altogether seven years running. In France, the wine harvest was late and small from 1591 to 1597. Grain crops fared even worse. . . . W. G. Hoskins observed, "the 1594 harvest [in England] was bad; 1595 was even worse; 1596 was a disaster; 1597 was bad too." A season of scarcity grew into a massive famine that was called the "great dearth." There were terrible scenes of suffering in many parts of Europe (Fischer 1996: 93; see also Lamb 1977: 461–466; Martín-Vide and Barriendos-Vallvé 1995: 214–217).

Nor were these problems limited to western Europe. In Russia, for example, "severe weather" is said to have led to famine conditions in 1583, 1585, 1587–89, 1591–92, and 1598 (Borisenkov and Pasetskii 1983: 196–199). Then, in 1601, a combination of cool summer temperatures, unusually heavy rainfall in some regions, and very dry conditions elsewhere

> brought disaster to the [Russian] people. The crops failed again in 1602 and also, to a considerable extent, in 1603. Famine reached catastrophic proportions. . . . It has been estimated that more than 100,000 people perished in [Moscow] alone. Starving people devoured grass, bark, cadavers of animals, and, on occasion, even other human beings. Large bands of desperate men that roamed and looted the countryside and sometimes gave battle to regular troops appeared at this time and became a characteristic phenomenon of the Time of Troubles [c.1598–1613] (Riasanovsky 1963: 178; see also Borisenkov and Pasetskii 1983: 196–199; and Crummey 1987: 216).

Climatic and economic conditions in the Mediterranean basin also deteriorated during this period. In addition to a string of "crisis years" in southern Europe that began around 1584, the eastern Mediterranean also experienced severe food shortages. Between 1585 and 1590, for example, parts of the Ottoman Empire "suffered every kind of evil: poverty, high prices, spectacular famines, and finally plague" (Braudel 1975, I: 593; see also ibid., pp. 80, 245–246, 272–273, 584; Griswold 1983; Inalcik 1994: 184, 441; Utterström 1955: 26, 37, 40; Glamann 1974: 456; Minchinton 1974: 117; Schöfer 1997; Elliott 1970: 298; Vilar 1974: 101; Martín-Vide and Barriendos-Vallvé 1995: 214–217). The situation in Anatolia and southeastern Europe was made even worse by a series of anti-government uprisings known as the Celali Revolts, which began in the 1590s and continued off and on until they finally were put down by Ottoman forces in 1610. These revolts appear to have been caused by a combination of factors including official corruption, the government's inability to find alternative employment for

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its irregular soldiers, and, according to some specialists, to the sharp fluctuations in climatic conditions that occurred in the eastern Mediterranean region at this time (Griswold 1983; Grove and Conterio 1995: 231–236).

Although our information for North America, Africa, South Asia, and Southeast Asia at this time is much less complete than for many other parts of the world, anomalous climatic conditions affected many of those regions as well. For example, tree-ring data indicate that the "Lost Colony" established by Sir Walter Raleigh in North Carolina during the summer of 1587 disappeared during that region's worst three-year drought (1587–89) in the last 800 years. Moreover, the same data indicate that the near collapse of the colony at Jamestown occurred during "the driest 7-year period" (1606–12) in that area in the last 770 years (Stahle et al. 1998: 564). Interestingly, a severe drought also struck the North Platte region of western Nebraska in 1587 and continued, with a respite in 1594–96, until 1605 (Wedel 1986: 45). In West Africa, the collapse of the Songhay Empire in 1591 also has been associated with drought (Lamb 1977: 471, n. 2), although there apparently were unusually heavy rains and severe flooding in and around the city of Timbuktu during the winters of 1592–93 and 1602–03 (Lamb 1982: 226). Records from Northeast Africa that are both incomplete and difficult to interpret indicate that water levels in the Nile River Valley generally were low between 1587 and 1623 and exceptionally low in 1588, 1601–02, 1613, and 1617–21. This may help to explain the serious economic problems that afflicted Egypt and the western sections of the Ottoman Empire during those years as well as the mass migrations in Ethiopia and the region near Lake Victoria that, drawing on local oral traditions as well as scientific data, some scholars now date to the late sixteenth or early seventeenth centuries (Webster 1979; Herring 1979: 59–62).

In parts of Northern India, the failure of the summer monsoon in 1596, which probably was related to the "moderate" warm ENSO event that occurred that year in the Indo-Pacific Basin (Quinn 1992: 123), led to a famine so severe that people are said to have engaged in cannibalism in a desperate attempt to stay alive (Moreland 1962: 119). Just a few years earlier, the great Mughal leader Akbar (1542–1605) had abandoned his new capital at Fatehpur Sikri, a development that many scholars have attributed, at least in part, to severe water shortages in the region during the 1580s (Lamb 1977: 471, n. 2; Lamb 1982: 227; Bryson and Murray 1977: 104–105; Parker 1979: 21; Quinn 1992: 123). Anthony Reid's research suggests that there were major food shortages in Vietnam in 1586, 1588–89, 1592, and every year between

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1594 and 1597. He also finds unusually dry conditions in parts of the Indonesian archipelago in 1603 and in many of the years between 1605 and 1616. Moreover, Dutch and Malay records indicate that 1606–08 was a time of severe famine in northwestern Sumatra (Reid 1997: 212–217). As was noted earlier, many of these years (1589–91, 1596, 1600–01, 1604, 1607–08, for example) also witnessed significant warm ENSO events in the Indo-Pacific Basin (Quinn 1992: 123).

Even before turning to East Asian history, therefore, it seems clear that the years between the mid-1580s and 1610 were a time of climatic upset on a global scale. Records from Japan, Korea, and China provide further proof of this. For example, despite the generally positive economic picture usually painted for Japan at this time, drought and famine were reported in various parts of the country in 1584–85; incessant rains, flooding, and widespread poor harvests in 1596; a severe winter in 1597–98: serious food shortages in some areas in 1599; incessant rains, flooding, and widespread harvest failures in 1601; poor harvests in central Honshū in 1602; flooding in the Kyōto region in 1603; flooding in the Kantō and drought and poor harvests elsewhere in 1604; poor harvests in the Tōhoku and Kantō in 1605; and, finally, a series of bad years between 1606 and 1610 in which heavy rains and floods appear to have alternated with very dry conditions to create severe food shortages in many areas. Also of interest is the fact that between 1602 and 1613, Japan apparently saw no fewer than nine mild winters, a situation that in later times is known to have had an adverse effect on agricultural production during the year following the mild winter (Arakawa 1955b: 145; Arakawa and Usami 1985: 324–325).

Conditions in Korea at this time may have been even worse than those in Japan. Despite the disruptions to record keeping and record preservation caused by the massive Japanese invasions of 1592–93 and 1597–98, Korean sources report famine conditions in 1584, spring frosts and heavy rains in 1590, unusually heavy snows early in 1591, floods and a severe famine in 1593, famine in 1594, heavy rains and floods in 1605, unusually cold conditions during the early summer of 1607, and severe drought and famine in 1608 (Kojŏn Kanhaenghoe 1957, I: 127, 131, 149).

During this same period in China, there was drought in the Beijing region in the autumn and winter of 1584, famine in Huguang during the late summer of 1585, severe flooding in Jiangnan, Zhejiang, Huguang, Fujian, Yunnan, and Liaodong in the summer of 1586, and heavy rains and serious flooding in both northern and southern China during the summer of 1587. At various points during 1587, drought and famine were reported in areas north of the Yellow River. By the

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early months of 1588, the areas experiencing severe food shortages had widened to include parts of Shaanxi, Shanxi, Shandong, Henan, Zhejiang, and Jiangnan. In 1589, an extended period of severe drought began in southeastern China, with parts of Zhejiang, Huguang, Jiangxi, and Jiangnan being particularly hard hit (Dunstan 1975: 9). After a brief respite in 1590, the next few years saw heavy rains and floods damage agricultural production in many regions of the country (Satō 1993: 273–274; Song 1992: 163, 176, 198, 296, 311–312, 320, 323, 327, 352; Brook 1998: 191–192). By 1594, a year that also saw famine conditions in Korea, people in some parts of China were

> reduced to eating the bark of trees and the seeds of grass. When this did not satisfy their hunger they even tried to eat the excrement of wild geese. The situation became not only tragic but also alarming, and an official had it represented in a picture and presented the picture to the Wan-li emperor [r. 1572–1620]. . . . Similarly, a censor sent a sample of the excrement . . . to the monarch in order to arouse his pity for the poor who were reduced to seeking nourishment from that filth (Chan 1982: 234–235; see also Satō 1993: 275).

Although the epicenter of the 1594 famine is usually considered to have been north-central China, there also were very poor harvests that year in the southeastern provinces of Fujian and Guangdong (Satō 1993: 275).

During the autumn of 1596, a year during which heavy rains severely damaged crops in various parts of Japan, there were "persistent rains and severe floods" in one of China's key agricultural regions, northern Zhejiang province (Satō 1993: 275). It also was about this time that the following description of a serious flood in Jiangxi province was written:

> . . . Nan-ch'ang, Chi-lin, and other districts near the Yangtze [River] were all flooded. Banks burst and were washed away. Sometimes several members of the same family perished in the waters; houses, farms, animals, and property were, of course, destroyed. Later, although the flood had subsided, water from the lakes lingered for months. Thus the autumn crops were delayed and there was no hope of a harvest. No one had ever experienced such desolation as this . . . (Wang Jiazhen quoted in Chan 1982: 233).

Although economic conditions in China may not have been as bad as those in Russia or western Europe during the early seventeenth century, they were bad enough. For example, in 1601, a year which may have been the coldest summer in the northern hemisphere in the past

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six centuries and during which many parts of Japan experienced unusually heavy rains and severe harvest failures, an extended drought in northern China was followed by summer floods in both the north and the southeast of the country. In 1607, a year that saw heavy rains in central Japan and unusually cool temperatures in Korea, floods struck both central and northern China. Finally, in 1609 the Ming Empire was hit by an extended drought in Huguang, Sichuan, Henan, Shaanxi, and Shanxi and by heavy rains and floods in Fujian, Zhejiang, and Jiangxi. By the late spring of 1610, famine conditions were reported in parts of Bei Zhihli, Shandong, Shanxi, Henan, Shaanxi, Fujian, and Sichuan (Satō 1993: 278; Song 1992: 163, 176, 182, 185, 189, 202, 205, 288, 296, 312, 327, 333).

As conditions in China deteriorated during the late sixteenth and early seventeenth centuries, the Ming government faced a number of significant military challenges to its rule. Among the more notable of those were a major rebellion in Shandong in 1587, repeated attacks in the northwest by the so-called Ordos Mongols, raids by the "Eastern Mongols" in Manchuria, hostilities between Chinese and Burmese forces on the southwestern frontier, Vietnamese incursions along the southern border, a lengthy rebellion in central and western China, and the two massive Japanese invasions of Korea during the 1590s. It also was during this period that the Manchu leader Nurhaci began to organize his people into the powerful political and military force that would eventually conquer China during the mid-1640s. To what extent Nurhaci was spurred in his organizational efforts by climatic and economic problems in the Manchu homeland remains to be carefully studied, but the rise of the Manchus, like the rise of the Jurchens in the late eleventh and early twelfth centuries and that of the Mongols in the late twelfth and early thirteenth centuries, occurred at a time when ice-core, tree-ring, and other evidence indicate that volcanic activity was affecting temperatures and altering wind and precipitation patterns in much of the northern hemisphere.

C. 1636–1644

Although there is evidence to suggest that volcanic activity was affecting global climate in a significant way during the 1620s and early 1630s, that evidence becomes even stronger during the late 1630s and early 1640s. Not only does Lamb report volcanic dust veils over the northern hemisphere for each year between 1636 and 1644, with especially

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heavy veils in 1640–43 (Appendix A),[21] but research by Robock and Free has revealed strong sulfate acid spikes in two or more northern hemisphere ice cores for 1641–44 (Appendix C) and in two or more southern hemisphere ice cores for 1641–42 (Robock and Free 1996a). Moreover, three different types of tree-ring evidence indicate that at some point during the summers of 1640–43, unusually cool temperatures adversely affected tree-ring growth in various parts of western North America (LaMarche and Hirschboeck 1984: 122–125; Scuderi 1990b: 73; Briffa, Jones, and Schweingrüber 1992: 746–747). Additional research suggests that 1641, 1642, and 1643 were the third, twenty-eighth, and tenth coldest summers respectively in the northern hemisphere as a whole during the past 600 years (Briffa et al. 1998). Tree-ring data from southern Japan indicate that 1631–40 and 1641–50 were the coldest and the third coldest decades in that region in the twenty-one decades between 1511 and 1720 (Kitagawa and Matsumoto 1995; Kitagawa 1999). Finally, the winters of 1641–42 and 1642–43 brought exceptional cold and snow to many parts of Japan (Arakawa 1955b: 145).

Based on our still-incomplete chronology of global volcanism, the most likely sources for the dust veils, sulfate acid spikes, and tree-ring anomalies mentioned above include: 1) the "cataclysmic" (VEI 4) 1638 eruption of Indonesia's Mt. Raung; 2) the February 1640 VEI 4 eruption of Chile's Mt. Llaima; 3) the "severe" (VEI 3) February 1640 eruption of Chile's Mt. Villarica; 4) the July 1640 VEI 3 eruption of Japan's Mt. Haku; 5) the "paroxysmal" (VEI 5) 31 July 1640 eruption of Japan's Mt. Komagatake, an eruption that continued for more than two months; 6) the December 1640 VEI 3 eruption of Indonesia's Mt. Awu; 7) the 1641 VEI 4 eruption of Indonesia's Mt. Kelut; 8) the 1641 VEI 3 eruption of the Philippine's Mt. Taal; 9) the 1641 VEI 3 eruption of Ecuador's Mt. Tungurahua (VEI 3); 10) the massive January 1641 eruption that occurred either at Indonesia's Mt. Awu (VEI 5?) or at the Philippines's Mt. Parker (VEI 6); 11) the March 1643 VEI 3 eruption of Japan's Mt. Miyakejima; and 12) the April 1643 VEI 3 eruption of New Guinea's Mt. Karkar. With their extraordinary power, it is likely that the 1640 eruption at Mt. Komagatake and the 1641

[21] Meteorological information in Chinese and Korean sources that was not available to Professor Lamb also indicates that there were veils of volcanic aerosols over Northeast Asia each year between 1636 and 1644 (Zhang et al. 1974, II: 413; Kojŏn Kanhaenghoe 1957, I: 97).

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eruption at either Mt. Awu or Mt. Parker had the most significant impact on global climate over the next several years.

Given this dramatic increase in both the frequency and intensity of global volcanic activity, it will come as no surprise that climatic conditions in East Asia were greatly disturbed over the next few years. For example, in 1639, the year following the VEI 4 eruption at Indonesia's Mt. Raung and a year in which Lamb records a significant volcanic dust veil over the northern hemisphere, there were drought and famine in the Ryūkyū Islands, in the Kyōto region of Japan, in parts of Korea, and in much of northern and central China (Satō 1993: 283; Song 1992: 177; Dunstan 1975: 10; Kojŏn Kanhaenghoe 1957, I: 148; Murai 1979: 450; Endō 1982: 59). The seriousness of the situation in North China at this time perhaps is indicated by the fact that 1639 is regarded by some scholars to have been the second driest year in that region during the two centuries between 1500 and 1699 (Zhang and Lin 1992: 291).

Drought and famine continued to plague North China the following year. Indeed, Zhang and Lin not only believe that 1640 was the driest year there between 1500 and 1699 but also was the single driest year in that region during the five centuries between 1470 and 1969 (Zhang and Lin 1992: 291). There are indications, however, that during 1640, weather patterns across China and East Asia were beginning to change from those that had prevailed over the previous several years. For example, during the late spring and early summer, there was heavy and sustained rainfall near the Yangzi River in what is now south-central Anhui province. This was followed during the fifth lunar month (21 May–18 June 1640) by a late frost that damaged crops in parts of modern Gansu province and in June and July by torrential rains and floods in parts of Zhejiang province in the southeast of the country (Satō 1993: 284; Song 1992: 313). These developments perhaps take on new significance when it is considered that the first seven months of 1640 witnessed at least four powerful volcanic eruptions around the world, including two in East Asia (Simkin and Siebert 1994: 194; Appendix B). Moreover, those eruptions were followed in 1640 by: 1) very poor harvests in northern Japan caused by exceptionally cool summer temperatures (Endō 1982: 60; Takahashi and Nemoto 1978: 184, Yamamoto 1989: 199); 2) unusual cold and snow in Korea during the early autumn (Yi 1995: 29); and 3) a very poor growing season in the mountains of the western United States, a situation that probably also was caused by unseasonably cool temperatures during the summer or early autumn (LaMarche and Hirschboeck 1984: 124; Scuderi 1990b: 73). It is therefore likely that during the summer and early autumn of 1640,

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northeast Asia, the northern Pacific, and western North America all were experiencing significant "anomalies" in atmospheric circulation that not only were shifting rain belts away from their "normal" positions but also were allowing cold Arctic air to penetrate much further south than ordinarily would have been the case (Lamb 1982: 34–35; LaMarche and Hirschboeck 1984: 124).

The harvest failures in 1640 in northeastern Japan, a region that over the previous century had become one of that country's key grain-producing regions (Arakawa 1979: 19–24; Arakawa 1982: 32–34), soon had serious effects elsewhere (Nakajima 1976: 8; Nagakura 1982: 73; Yamamoto 1989: 199; Totman 1993: 111–112). The year 1640 also brought severe problems to China where, from the Huai Valley to the Northern Metropolitan Region, "all the bark had been stripped from the trees and people [even] dug up corpses for food" (Zhang et al. 1974, II: 511; see also Satō 1993: 283–284; Song 1992: 177; Dunstan 1975: 10). Another contemporary source reports that following a rebellion in Shandong during which a large number of people were killed, the dead quickly were cut up for food (Chan 1982: 235). Economic conditions remained terrible in 1641. Not only did unusually heavy snowfalls and food shortages cause serious problems in southeastern China during the early months of the year, it remained very dry in much of the north with infestations of locusts being reported in many areas (Song 1992: 161, 177, 459). Moreover, Zhang and Lin believe that 1641 not only was dry in northern China but also was the second driest year in *central* China (27°–35° N) in the two centuries between 1500 and 1699 (Zhang and Lin 1992: 291). In any event, drought conditions in western Shandong were so severe in 1641 that the Grand Canal apparently dried up in the vicinity of Linqing (Satō 1993: 284). As grain prices rose to extraordinary levels, local people are said to have organized bandit gangs and even resorted to cannibalism to stay alive (Zhang et al. 1974, II: 511–512). People elsewhere tried to avoid starvation in other ways:

> In the mountains near Yenqing Temple, the people of Jintan [in the southern portion of modern Jiangsu province] saw a person who told them: '[I]f you dig in the earth here down to a depth of more than a foot then you will find soil that is edible.' They did as he said, washing and pounding the earth to make 'flour' and 'gruel' which they ate. Those who [did this] grew more numerous every day. Similarly . . . , [in nearby] Changshan people also dug up some earth that was fit to eat. It was greenish-white in color like the tuckahoe root. Similarly, in [nearby] Shizi-jian the soil was yellowish-red in color, had the appearance of pork liver, and was commonly called 'Guanyin flour'.

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> Many of those who ate it suffered abdominal pains that caused them to fall to the ground; in the end they lay scattered about [like casualties of war] and died (Zhang et al. 1974, II: 511).

During the early spring of 1641, just a few weeks after special prayers for rain had been invoked at the court in Beijing (Satō 1993: 284), severe drought also was reported in nearby Korea, a situation that apparently persisted into the summer (Kojŏn Kanhaenghoe 1957, I: 148). That same year also brought adverse weather conditions to Japan. For example, during the summer when dwindling grain supplies, continuing drought, and unusually cool temperatures were driving up grain prices in parts of China and Korea, the same was true on the Japanese island of Shikoku, in portions of central Honshū, and perhaps in Kyūshū as well (Endō 1982: 61; Yamamoto 1989: 199; Kitagawa and Matsumoto 1995; Kitagawa 1999). Moreover, the situation in Japan was made even worse by the fact that incessant rains and unusual cold were bringing the second poor harvest in a row to the agriculturally important northeast (Endō 1982: 61; Nakajima 1976: 7–9; Takahashi and Nemoto 1978: 184; Yamamoto 1989: 199).[22] A few months later Tokugawa authorities were speaking openly of the need to combat a "nationwide famine" *(tenka no kikin)* (*Tokugawa jikki* 1959–67, XL: 258; Endō 1982: 62; Nagakura 1982: 75–76; Totman 1993: 110–113).

Unfortunately, the winter of 1641–42 was very cold in Japan and the summer of 1642 was very cool in the northern hemisphere as a whole, all of which may help to explain the poor harvests that once again afflicted northeastern Japan (Arakawa 1955b: 145; Takahashi

[22] It is possible that the "paroxysmal" (VEI 5) eruption of Mt. Komagatake in southern Hokkaidō that began on 31 July 1640 and lasted until 9 October of that year not only had a direct impact on agricultural yields for 1640 in northern Honshū but also on the yields for 1641 and 1642 as well. Not only did this eruption occur quite near the agriculturally important Tōhoku region, but the strong sulphate-acid spikes recorded in northern hemisphere ice cores for 1641, 1642, and 1643 (Appendix C) indicate that the eruption at Mt. Komagatake and/or eruptions elsewhere in the world during these years were "sulfur-rich" and thus were likely to have had a significant impact on global and regional climate. Given the latitude (42.07° N) of Mt. Komagatake and the prevailing wind patterns, it is likely that the volcanic aerosol it produced blanketed much of northern Japan, northern Korea, Manchuria, parts of northern China, Mongolia, and other northern parts of the hemisphere before gradually drifting poleward and dispersing. Moreover, as that aerosol did disperse, it may have been replaced by another "sulfur-rich" aerosol produced by the "colossal" (VEI 6) eruption that occurred either at Indonesia's Mt. Awu or the Philippines's Mt. Parker in January 1641. In any event, research on tree-ring densities indicates that the summer of 1641 was the third coldest (after 1601 and 1816) in the northern hemisphere as a whole over the past 600 years. Over those same six centuries, 1642 and 1643 ranked as the twenty-eighth and tenth coldest summers respectively (Briffa et al. 1998: 450).

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and Nemoto 1978: 284). Elsewhere in the country, heavy rains and floods reduced agricultural production in the Kantō and in parts of Kyūshū (Yamamoto 1989: 199). Still other areas were forced to contend with severe drought conditions (Nakajima 1976: 7–9; Arakawa and Usami 1985: 324–325). The results were catastrophic. Yamamoto Hirofumi has estimated that from 1633 to 1643, between 500,000 and 1,000,000 Japanese may have died from famine-related causes, most during the peak years (1641–42) of what has come to be known as the "Great Famine of the Kan'ei Period" *(Kan'ei dai kikin)* (Yamamoto 1989: 197). A Japanese document from that period describes the situation in 1642, a time when much of the country apparently was still feeling the effects of several years of poor harvests and the exceptionally cold winter of 1641–42:

> From [the second] to the fifth month [1 March–27 May 1642], there was a serious famine throughout the country. The corpses of those who had starved to death filled the streets while the peasants, artisans, and merchants who begged for food were numerous. None of these people had any clothes [and could only] . . . wrap straw and other matting around their bodies for protection. They begged in the marketplace and slept along the side of the road (*Kan'ei nikki zōho* quoted in Asao 1975: 368; see also *Tokugawa jikki* 1959–67, XL: 258; Arakawa 1955b: 145; Endō 1982: 62; Vlastos 1986: 35–36).

Nor did warmer temperatures later in the year do much to alleviate the situation. For example, in August the authorities in Edo were informed that grain shortages and soaring prices were continuing to cause severe hardships in many parts of the country (*Tokugawa jikki* 1959–67, XL: 280).

At some point during that same year, a native of China's Yangzi delta, a region which until a few years earlier had been one of the richest on earth, wrote the following:

> Formerly, when I was told of cannibalism in Shantung and Honan, I was inclined to be skeptical. But now this practice occurs frequently inside and outside the city of Su-chou. In spite of severe penalties imposed by the authorities for this offence, it often is committed. In the streets there are numerous beggars, very thin and worn. *Moreover, since the new year [31 January 1642], it has been cold and it has rained frequently. The spring has nearly come to an end, but the cold still persists. After the full moon of the second month [15 March], it rained continuously for over ten days.* The people are dying in great numbers through lack of food. I have seen with my own eyes several tens of (starved) corpses being buried daily in the property of the prince. When the price of rice

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> rises to over ninety coins a pint, what wonder if they have nothing to eat! Most of the residences in the city are empty and they are falling into ruins. Fertile farms and beautiful estates are for sale but there is no one to buy them. Formerly the city of Su-chou was prosperous and its people tended to be extravagant. It is natural that after a period of prosperity a period of depression should follow; but I never dreamed that I should have to witness these misfortunes in the days of my life (Ye 1911: 2/10b, quoted in Chan 1982: 235–236 [my emphasis]).

Nor were China's problems limited to the lower Yangzi region. There was severe flooding in Henan during the summer of 1642 (Song 1992: 297) as well as famine in portions of Henan, Shandong, Huguang, Zhejiang, and the Northern Metropolitan Region. Ominously for Chinese authorities, 1642 also saw the final collapse of Ming military power north of the Great Wall and an extended campaign by Manchu forces in eastern China. To what extent the Manchus were experiencing climatic disasters of their own remains to be thoroughly studied, but, as was the case during the late sixteenth century, there is considerable evidence to suggest that their attacks on China during the 1630s had been inspired, at least in part, by serious economic problems in the Manchu homeland (Roth 1979: 27; Wakeman 1985: 58; Marks 1998: 140).

Lamb's Dust Veil Index, sulfate acid spikes in various ice cores, tree-ring research, and meteorological observations in Chinese sources all indicate that wind and precipitation patterns in the northern hemisphere continued to be affected by volcanic aerosols in the atmosphere for the next several years. The winter of 1642–43 again was very cold in Japan and the summer of 1643, like those of 1641 and 1642, was very cold, perhaps the tenth coldest in the northern hemisphere as a whole in the past 600 years. Precisely what all of this meant for China in 1643 remains to be carefully studied, but drought was a serious enough problem in some regions for the Ming emperor once again to order special prayers for rain (Zhang 1974, II: 455, 486; Song 1992: 177). Severe drought also was reported that year on the Japanese island of Shikoku, in the Ryūkyū Islands in the East China Sea, on the island of Taiwan, and in the northern Philippines (Arakawa and Usami 1985: 326; Lamb 1982: 227; Endō 1982: 59; Blair and Robinson 1903–09, XXXV: 123).

Although the climatic problems of the late 1630s and early 1640s were particularly severe in East Asia, many other parts of the northern hemisphere were affected as well. For example, in the Ukraine drought and famine are recorded in 1637, 1638, and 1639, and there apparently was an exceptionally cold winter in that region in 1638–39 (Borisen-

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kov and Pasetskii 1983: 208–210). Very dry conditions apparently also prevailed in Russia and the Ukraine during these years, with no fewer than eleven droughts being recorded between 1640 and 1659 (Borisenkov 1995: 174–175). In South Asia, there were droughts in Madras in 1640–41, in northern and northwestern India in 1641, along the east-central coast of India in 1642, and in Bengal during the winter of 1642–43 (Moreland 1990: 208). In addition to the six major volcanic eruptions that occurred in Indonesia, New Guinea, and the Philippines between 1638 and 1643, tree-ring and other data indicate there was a severe drought in Java in 1637 and a notable "drought-induced famine" in Vietnam during the warm ENSO year of 1641 (Reid 1997: 212–214).

Further to the west, there were very low water levels in the Nile Valley in 1641 (Herring 1979: 83), a situation that correlates well with what is known about teleconnections between climatic conditions in eastern Africa and Japan (Yamamoto 1978: 61; Manabe 1997). In the eastern Mediterranean, exceptionally stormy weather during the winter of 1637–38 was followed by heavy snows in the winter of 1638–39, drought conditions in the spring of 1639, and a major drought in Crete and the Greek Archipelago in 1643–44 (Grove and Conterio 1995: 235–239). In France, there was a series of cold, wet summers between 1639 and 1643 that "proved disastrous to grain production" and led to serious food shortages in many parts of the country (Ladurie 1972: 151; Coveney 1977: 27). In England, the period from 1620 to 1650 was "among the most terrible years through which the country has ever passed. It is probably no coincidence that the first real beginnings in the [English] colonization of America date from this period" (Bowden 1967: 621). Unfortunately for some colonists, climatic conditions in North America appear to have been little better than those they had left behind. For example, tree-ring studies from southeastern Virginia show sharp fluctuations in precipitation throughout the late 1630s and early 1640s. Indeed, while 1638 may have been the single driest year in that region during the entire seventeenth century, the early 1640s were wet and cool (Stahle 1998: 565). In New England, a severe winter in 1637–38 was followed by exceptional storminess during the summer and early autumn of 1638, a cool wet summer in 1641, and another severe winter in 1641–42. Concerning that last winter, a time when parts of southeastern China and much of Japan also were experiencing unusually cold conditions (Chan 1982: 235–236; Arakawa 1955b: 145), the governor of the Massachusetts Bay Colony, John Winthrop, wrote the following:

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> The frost was so great and continual this winter that all of [Massachusetts Bay] was frozen over, so much so and so long, as the like, by the Indians' relation, had not been so these forty years, and it continued from [28 January to 3 March 1642], so as horses and carts went over in many places where ships have sailed. . . . To the southward also the frost was as great and the snow as deep, and at Virginia itself the great [Chesapeake] bay was much of it frozen over, and all of their great rivers, so as they lost much cattle for want of hay, and most of their swine (Winthrop 1996: 384; see also Ludlum 1966: 15, 33).

The above passage perhaps takes on new significance when it is considered that yet another notable winter in eastern North America (1783–84), during which intense cold paralyzed parts of New England and sections of Chesapeake Bay again froze, also occurred just after a series of powerful and sulfur-rich eruptions in various parts of the world. Moreover, as had been the case during the late 1630s and early 1640s, those eruptions and that cold winter coincided during the 1780s with serious economic and political difficulties in Japan, Korea, China, India, Northeast Africa, and western Europe (Lamb 1970: 509; Lamb 1982: 237–238; Totman 1993: 238; Arakawa 1957b: 211–217; Arakawa 1979: 81–175; Arakawa and Usami 1985: 184; Satō 1993: 366; Mikami 1987; Ladurie 1971: 71–74; Simkin and Siebert 1994: 202; Mooley and Pant 1981: 471; Evans 1994: 46; Issawi 1982: 98; Parker and Smith 1997; Marks 1998: 215–217).

Conclusion

Some time ago, I suggested that an important task for those interested in the global dimensions of the so-called "general crisis of the seventeenth century" might be a narrowing of chronological focus. A better understanding of "what went wrong" in various parts of the world between 1630 and 1650, I argued, might "go a long way towards helping us understand not only some of the important issues in mid-seventeenth-century history but in pre-modern history in general" (Atwell 1997: 247). As the graphs in the appendixes of this article indicate, among the many things that "went wrong" during the 1630s and 1640s were sharp fluctuations in global temperatures which, along with significant changes in wind and precipitation patterns, had a devastating impact on agricultural production in many parts of the world. Moreover, those graphs suggest that in terms of the history of climate at least, instead of one long "seventeenth-century crisis," there may have been four or five shorter ones during which anomalous climatic condi-

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tions coincided with notable volcanic activity. Nor was the seventeenth century unique in this regard. Data from various sources indicate that over the centuries between 500 and 1899, sulfur-rich volcanism also may have had a significant impact on climate in the northern hemisphere during the following periods: c. 535–36, 564–601, 622–28, 640–42, 702–711, 750–57, 820–35, 870–74, 934–36, 959–61, 1003–09, 1018–30, 1068–74, 1099–1106, 1118–28, 1166–79, 1285–1300, 1330–53, 1485–99, 1537–44, 1664–80, 1693–1702, 1730–34, 1740–44, 1752–56, 1781–85, 1810–19, 1828–43, 1861–67, and 1884–92. Many of these periods have long been considered times of significant economic and political change in East Asian and world history.

This, of course, should not be interpreted to mean that the study of volcanism and climate is the key that will unlock the mysteries of pre-modern history. Even in the study of short-term climatic change, it is clear that sulfur-rich volcanism is just one of many variables that must be taken into account. Nevertheless, as was noted in Part One above, the climatic anomalies and particularly the declines in summer temperatures associated with volcanic eruptions often have had "catastrophic consequences for the world's food supply" (American Geophysical Union 1992: 3). Given that fact, it might be well for scholars to keep volcanic activity in mind as they consider

> the congenital frailty of man compared to the colossal forces of nature. . . . Historians of the *ancien regime* are quite right to regard it as punctuated by the succession of good, not so good, or bad harvests. These were the regular drumbeats which set in motion enormous fluctuations of prices on which so many other things depended. And who could fail to agree that this insistent background music was in part determined by the changing history of climate (Braudel 1981: 50–51).

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Appendix A

Dust Veil Years over the Northern Hemisphere, 1500–1699

Year	DVI*	Year	DVI*	Year	DVI*
1500†	**200**	1606	20	1646	60
1501	**150**	1607	50	1647	45
1502	**100**	1608	50	1648	30
1503	50	1609	50	1649	15
1535	50	1610	40	**1650**	**100**
1536	50	1611	30	1651	75
1537	50	1612	20	1652	50
1553	**100**	1613	10	1653	25
1554	**500**	**1614**	**200**	**1660**	**340**
1555	**350**	**1615**	**150**	**1661**	**255**
1556	**200**	**1616**	**100**	**1662**	**170**
1557	**100**	1617	50	1663	85
1586	**200**	**1625**	**100**	**1664**	**130**
1587	**150**	1626	75	**1665**	**100**
158	**100**	1627	50	1666	65
1589	50	1628	25	1667	30
1593	**200**	**1631**	**120**	**1672**	**200**
1594	**150**	1632	90	**1673**	**150**
1595	**100**	1633	60	**1674**	**100**
1596	50	1634	30	1675	50
1597	40	1636	40	**1680**	**280**
1598	30	1637	30	**1681**	**210**
1599	20	**1638**	**120**	**1682**	**140**
1600	10	1639	85	1683	70
1601	**400**	**1640**	**150**	**1693**	**140**
1602	**300**	**1641**	**400**	**1694**	**285**
1603	**210**	**1642**	**275**	**1695**	**205**
1604	**110**	**1643**	**175**	**1696**	**105**
1605	10	1644	75	1697	45

Source: Lamb 1970: 526–527

*For technical discussions of the ways in which Professor Lamb arrived at his DVI estimates, see Lamb 1970: 460–476. Note that according to Professor Lamb's calculations, the single heaviest volcanic dust veil in the northern hemisphere between 1500 and 1966 occurred in 1815, the year of a powerful eruption at Indonesia's Mt. Tambora. Professor Lamb rated the dust veil for that year at 695. For comparative purposes, it might be noted that he rated the dust veil for 1883 (following the eruption of Mt. Krakatau) at 400 and the dust veil for 1963 (following the eruption of Mt. Agung) at 160.

†**Bold type** indicates years with DVI ratings ≥ 100.

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Appendix B

Years in Which There Were One or More Confirmed Eruptions* Rated at VEI 4 or Greater, 1200–1699

Year	Number of Eruptions and VEI Ratings	Year	Number of Eruptions and VEI Ratings
1267	1 (VEI 4)	1595	1 (VEI 4)
1306	1 (VEI 4)	1597	1 (VEI 4)
1311	1 (VEI 4?)†	1600	2 (VEI 4, 6)
1357	1 (VEI 4)	1605	1 (VEI 4)
1362	1 (VEI 5)	1606	1 (VEI 4)
1416	1 (VEI 4?)	1622	1 (VEI 4)
1444	1 (VEI 4)	1625	1 (VEI 4)
1471	1 (VEI 5?)	1630	1 (VEI 4)
1477	1 (VEI 5?)	1631	1 (VEI 4)
1480	1 (VEI 5+)‡	1638	1 (VEI 4?)
1482	1 (VEI 5)	1640	2 (VEI 4, 5)
1510	1 (VEI 4)	1641	2 (VEI 4, 6)§
1534	1 (VEI 4)	1646	1 (VEI 4?)
1552	1 (VEI 4)	1650	1 (VEI 4?)
1563	1 (VEI 4)	1660	2 (VEI 4, 4)
1580	1 (VEI 4)	1663	1 (VEI 5)
1581	1 (VEI 4)	1667	1 (VEI 5)
1582	1 (VEI 4)	1673	1 (VEI 5?)
1585	1 (VEI 4)	1680	1 (VEI 5?)
1586	1 (VEI 5?)	1693	2 (VEI 4, 4?)
1593	1 (VEI 5?)	1694	1 (VEI 4)

Source: Simkin and Siebert 1994: 184–251

*Although carbon-14 dating, tephrochronological studies, ice-core studies, tree-ring studies, and historical accounts have confirmed a number of additional eruptions rated at VEI 4–7 between 1200 and 1699, scientists have not been able to date those eruptions to specific years. Among the more notable of those eruptions are: 1) Okataina (New Zealand), VEI 5, 1180 ± 20; 2) Unknown Source, VEI ?, 1259 ± 10; 3) Unknown Source, VEI ?, 1279 ± 10; 4) Kuwae (SW Pacific), VEI 6, 1452 ± 10; 5) Billy Mitchell (SW Pacific), VEI 6, 1580 ± 20; and 6) Long Island (New Guinea), VEI 6, 1660 ± 20.

†A ? following a VEI rating indicates that the rating was particularly difficult to assign or that the rating was based on circumstantial evidence.

‡A + following a VEI rating indicates an eruption volume in the upper third of the range of that particular VEI designation.

§Recent research indicates that the 3 January 1641 eruption that formerly was ascribed to Indonesia's Mt. Awu and rated at VEI 5 actually may have occurred at the Philippine's Mt. Parker and should be rated at VEI 6.

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Appendix C

Northern Hemisphere Ice-Volcanic Index, 1200–1699*

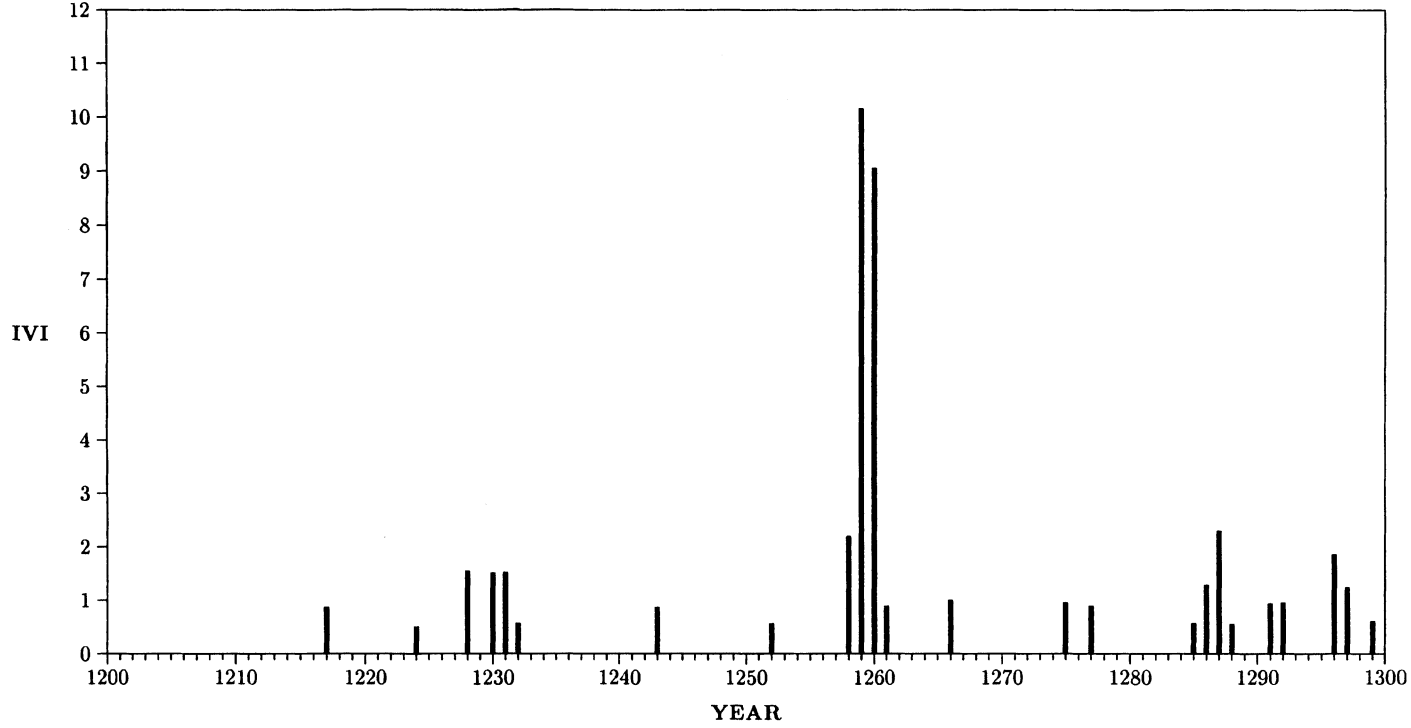


FIGURE C1. NH IVI 1200–1299.

*The IVI numbers represent the average sulfate ion concentrations or acidity in two or more northern hemisphere ice cores, depending on the years in question. For technical details, see Robock and Free 1995, 1996b.

Source: Robock and Free 1996a.

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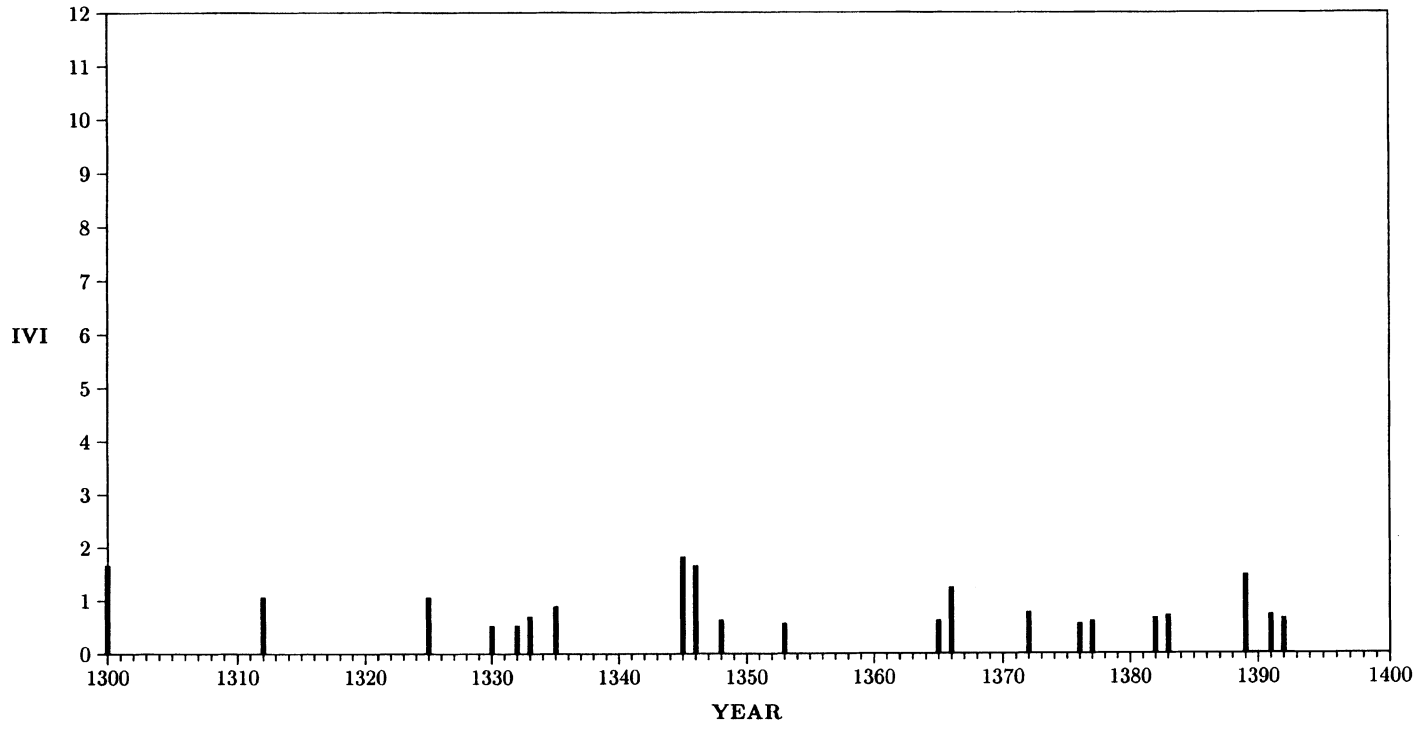


FIGURE C2. Northern hemisphere ice-volcanic index 1300–1399.

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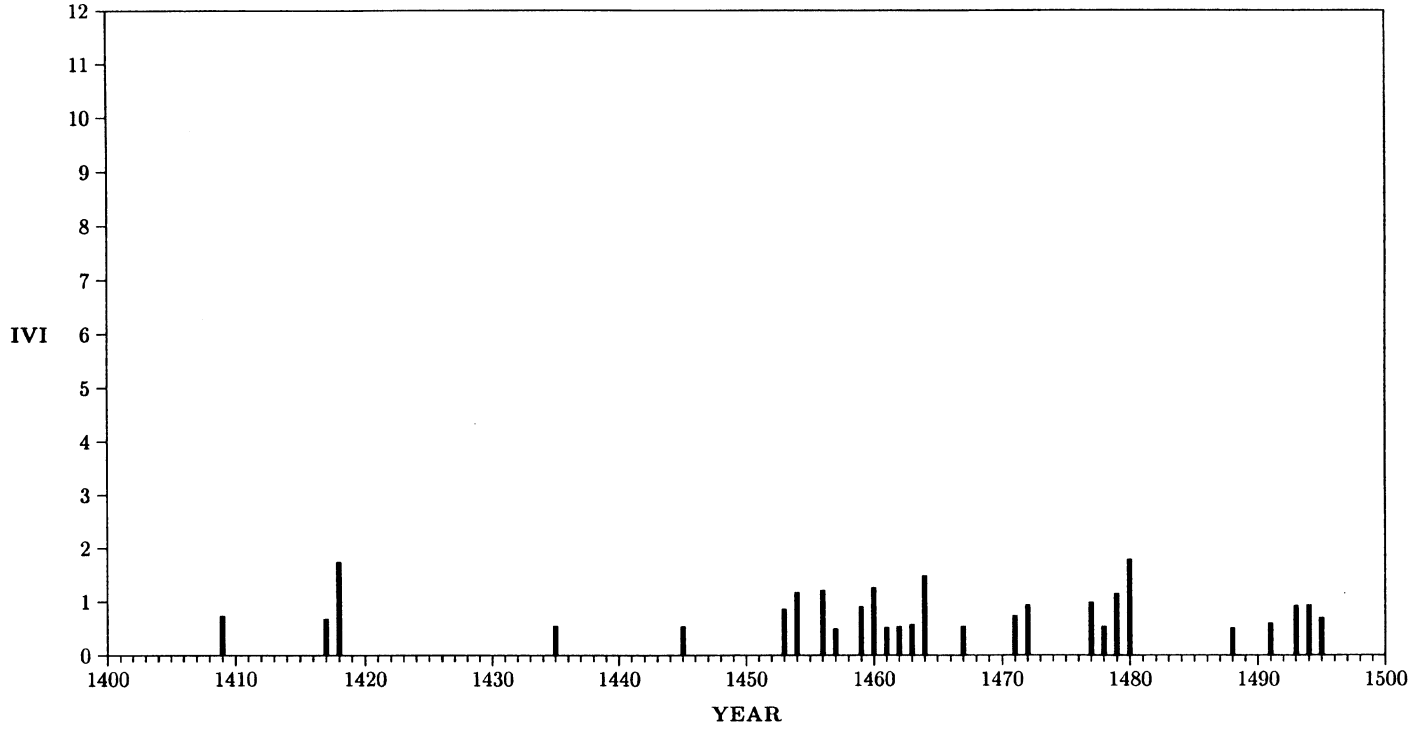


FIGURE C3. Northern hemisphere ice-volcanic index 1400–1499.

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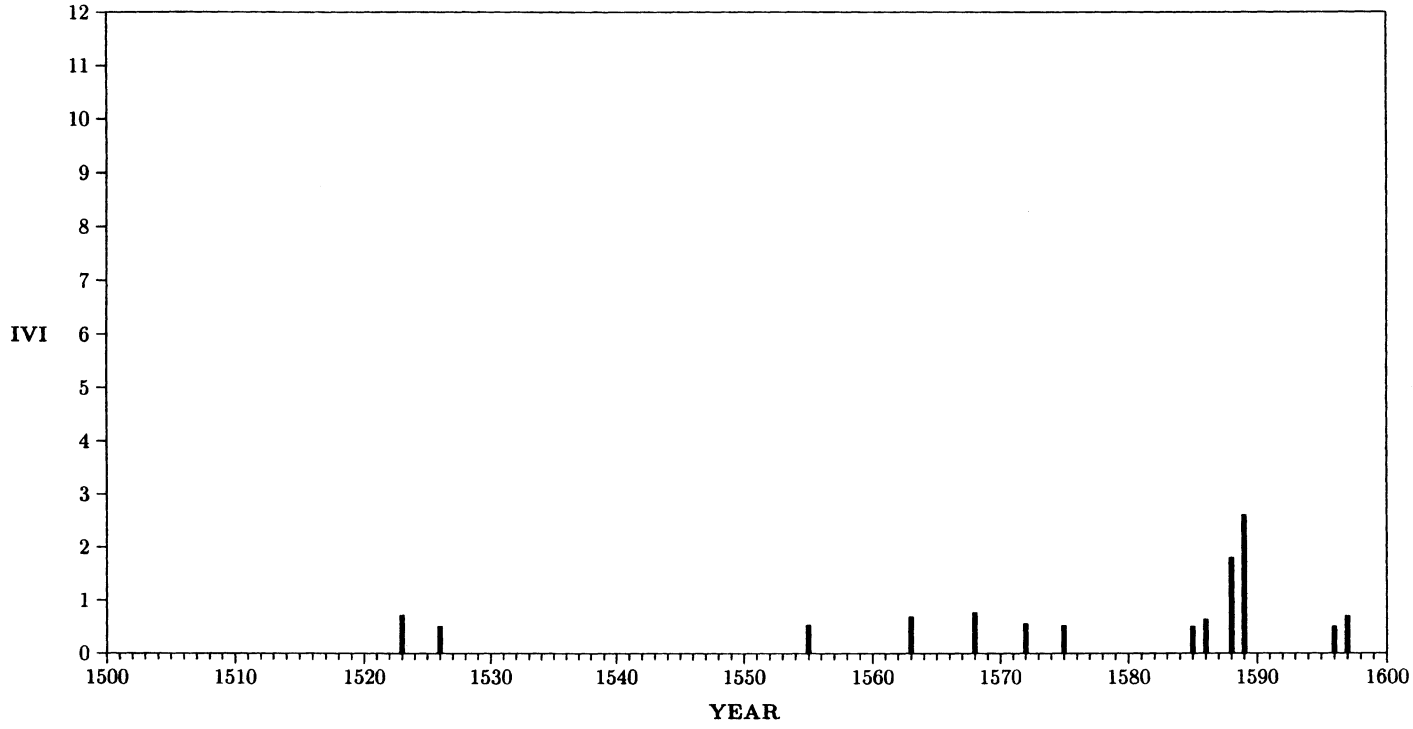


FIGURE C4. Northern hemisphere ice-volcanic index 1500–1599.

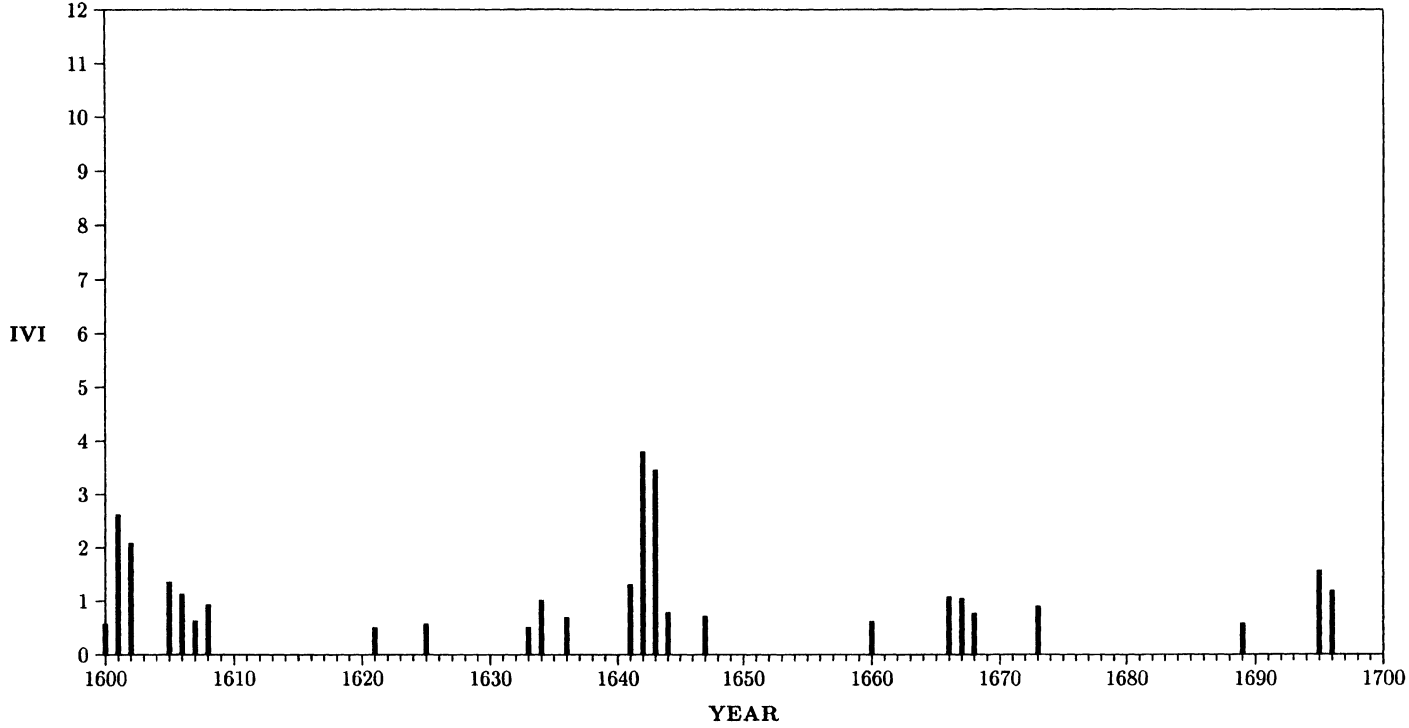


FIGURE C5. Northern hemisphere ice-volcanic index 1600–1699.

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Appendix D

Years in Which There Was Significant Frost Damage and/or Extremely Narrow Tree-Ring Growth in the Mountains of the Western United States, 1200–1699

1200	1299	1601	1666
1224	1453	1605	1680
1227	1485	1640	
1257	1500	1660	

Sources: LaMarche and Hirschboeck 1984: 122–125; Scuderi 1990b: 73; Bradley and Jones 1995b: 617

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Appendix E

Estimated Northern Hemisphere Summer Temperature Anomalies, 1400–1699*

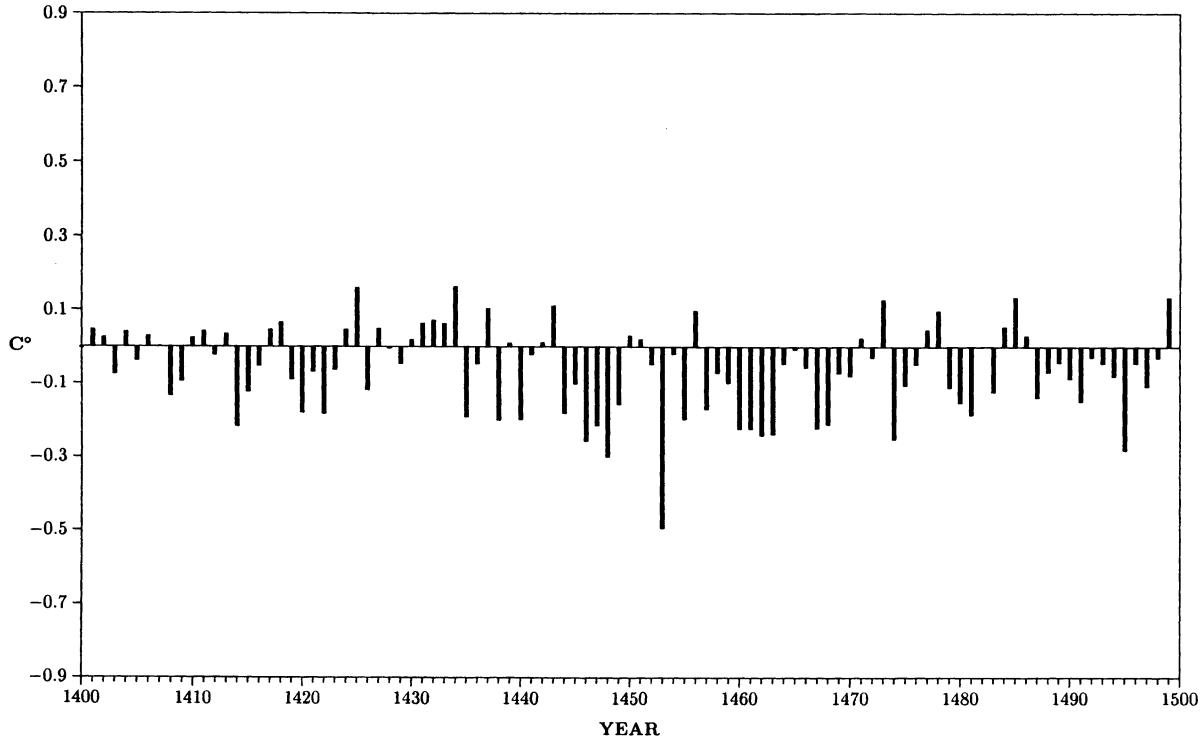


FIGURE E1. Estimated northern hemisphere summer temperature anomalies, 1400–1499.

*These anomalies were calculated using summer (April–September) monthly mean temperatures for the period 1881–1960 as a base. For technical details, see Briffa et al. 1998: 454.

Source: Briffa et al. 1998.

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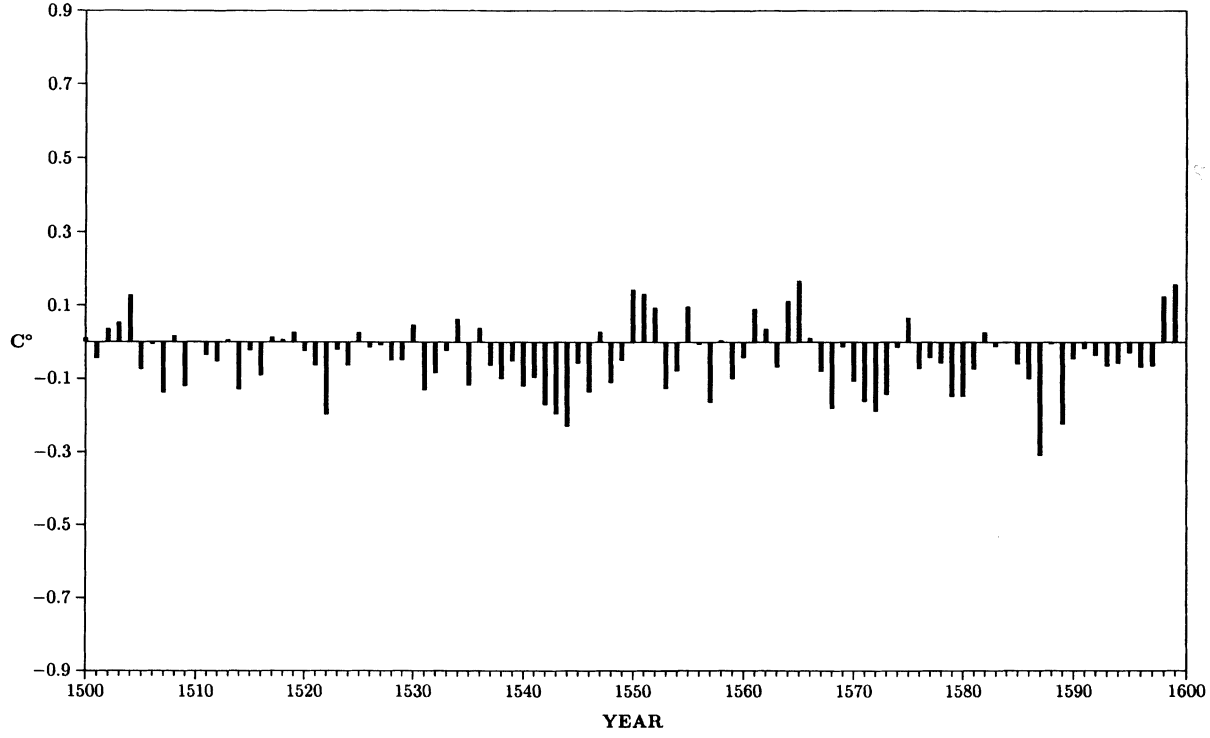


FIGURE E2. Estimated northern hemisphere summer temperature anomalies, 1500–1599.

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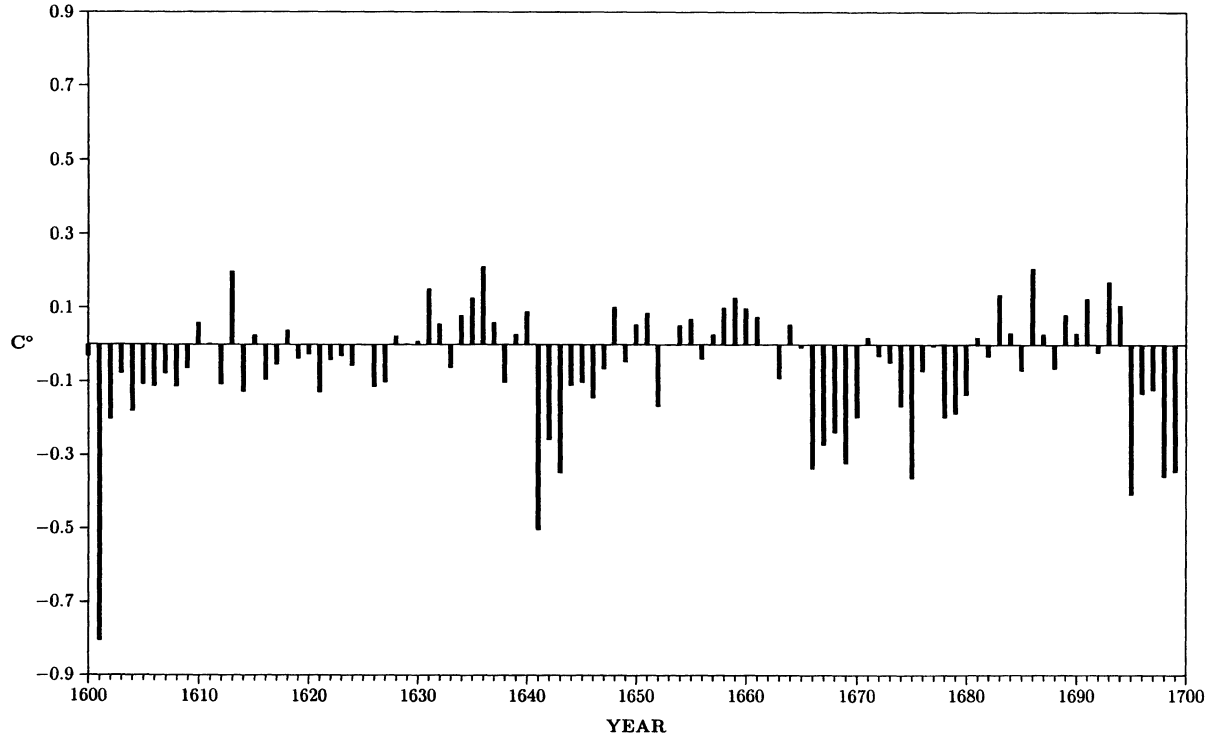


FIGURE E3. Estimated northern hemisphere summer temperature anomalies, 1600–1699.

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Appendix F

Incomplete List of Years in Which Dust-Veil, Ice-Core, and Tree-Ring Data Indicate that Volcanic Aerosols May Have Affected Climatic Conditions in the Northern Hemisphere, 1200–1699

Year	Nature of Evidence*	Year	Nature of Evidence*	Year	Nature of Evidence*
1200	TR	1376	IC	1495	IC, TR
1217	IC	1377	IC	1500	TR, DV
1224	IC, TR	1382	IC	1501	DV
1227	TR	1383	IC	1502	DV
1228	IC	1389	IC	1523	IC
1230	IC	1391	IC	1526	IC
1231	IC	1392	IC	1544	TR
1232	IC	1409	IC	1553	DV
1243	IC	1414	TR	1554	DV
1252	IC	1417	IC	1555	IC, DV
1257	TR	1418	IC	1556	DV
1258	IC	1435	IC	1557	DV
1259	IC	1438	TR	1563	IC
1260	IC	1445	IC	1568	IC
1261	IC	1446	TR	1572	IC
1266	IC	1447	TR	1575	IC
1275	IC	1448	TR	1585	IC
1277	IC	1453	IC, TR	1586	IC, DV
1285	IC	1454	IC	1587	TR, DV
1286	IC	1456	IC	1588	IC, DV
1287	IC	1457	IC	1589	IC, TR
1288	IC	1459	IC	1593	DV
1291	IC	1460	IC, TR	1594	DV
1292	IC	1461	IC, TR	1595	DV
1296	IC	1462	IC, TR	1596	IC
1297	IC	1463	IC, TR	1597	IC
1299	IC, TR	1464	IC	1600	IC
1300	IC	1467	IC, TR	1601	IC, TR, DV
1311	IC	1468	TR	1602	IC, TR, DV
1325	IC	1471	IC	1603	DV
1330	IC	1472	IC	1604	DV
1332	IC	1474	TR	1605	IC, TR
1333	IC	1477	IC	1606	IC
1335	IC	1478	IC	1607	IC
1345	IC	1479	IC	1608	IC
1346	IC	1480	IC	1612	IC
1348	IC	1485	TR	1614	DV
1353	IC	1486	IC	1615	DV
1365	IC	1491	IC	1616	DV
1366	IC	1493	IC	1621	IC, TR
1372	IC	1494	IC	1625	IC, DV

(Continued on next page)

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Appendix F *(continued)*

Incomplete List of Years in Which Dust-Veil, Ice-Core, and Tree-Ring Data Indicate that Volcanic Aerosols May Have Affected Climatic Conditions in the Northern Hemisphere, 1200–1699

Year	Nature of Evidence*	Year	Nature of Evidence*	Year	Nature of Evidence*
1631	DV	1660	IC, TR, DV	1675	TR
1633	IC	1661	DV	1680	TR, DV
1634	IC	1662	DV	1681	DV
1636	IC, TR	1664	DV	1682	DV
1638	DV	1665	DV	1686	TR
1640	TR, DV	1666	IC, TR	1689	IC
1641	IC, TR, DV	1667	IC, TR	1693	DV
1642	IC, TR, DV	1668	IC, TR	1694	DV
1643	IC, TR, DV	1669	TR	1695	IC, TR, DV
1644	IC	1672	DV	1696	IC, DV
1647	IC	1673	IC, DV	1698	TR
1650	DV	1674	DV	1699	TR

*DV = Dust veil index rating ≥ 100 (see Lamb 1970: 526–527)

IC = Ice-core evidence (see Appendix C)

TR = Tree-ring evidence (see LaMarche and Hirschboeck 1984: 122–125; Scuderi 1990b: 73; Briffa, Jones, and Schweingrüber 1992: 746–747; Briffa et al. 1998)

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